A Systematic Search for Ultra-Fast Outflows in Active Galactic Nuclei And Multi-Phase outflows study of the source SMSS J114447.77-430859.3

A thesis submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science (Research)

by

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Undergraduate Programme
Indian Institute of Science



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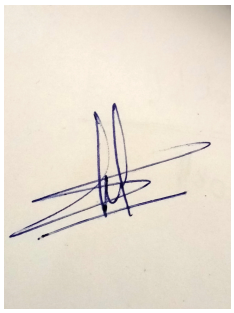
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Declaration

I certify that the work contained in this report has been done by me under the guidance of my supervisors. This work has not been submitted to any other institute for degree or diploma requirements. All the work contained herewith has been performed by conforming to the norms and guidelines given in the institute's Ethical Code of Conduct. I have cited all the relevant materials that have been used in this thesis.

April 12, 2026



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Certificate

This is to certify that the Bachelor's thesis titled "A Systematic search for Ultra-Fast Outflows in Active Galactic Nuclei and multi-Phase outflows study of source SMSS J114447.77-430859.3 " submitted by Ritik Kumar (Sr No: 11-01-0010-91-22-1-22209) to the Indian Institute of Science, Bangalore for the partial fulfillment of requirements for the award of the degree of Bachelor's of Science(Research) in Physics is a record of bona fide work carried out by him under my supervision and guidance during the academic year 2025-26.

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Abstract

In this thesis, we performed the sytematic survey of outflows detection in AGNs using the XMM-Newton data. In this work, we have done systematic X-ray spectroscopy study of 473 AGN from the Milli-Quas catalog using XMM-Newton EPIC-pn data to search for ultra-fast outflow signatures and other features like warm absorber, soft excess, and reflection. After studying all 473 sources, we have made categories based on the nature of each AGN — some showing warm absorber, some showing soft excess, some showing strong reflection, and some showing outflow evidence including multi-component outflows. From this, we have selected around 30 sources which were showing clear outflow signatures and detailed study of those has not been done yet. We have done the detailed spectral analysis of these 30 sources to characterize their outflow properties. From these 30 sources, we have picked one quasar — SMSS J114447.77430859.3 (J1144) at redshift z = 0.831 — which is the most luminous quasar in last 9 billion years and has not been studied in detail for outflows before. We have used two XMM-Newton observations of this source from 2023 and 2024 and found two outflow components moving at different velocities — a warm absorber at $v = 0.230c$ and an ultrafast outflow at v = 0.289 c — which are part of a single stratified disc wind. Both components are detected consistently in both observations separated by 6 months, and these outflows are energetically powerful enough to provide AGN feedback.

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Chapter 1

Introduction

1.1 Active Galactic Nuclei

Active galactic nuclei (AGN) are among the most energetic and luminous persistent sources in the Universe. They are powered by the accretion of matter onto a supermassive black hole (SMBH) residing at the centre of their host galaxies Lynden-Bell [1969]. The released gravitational potential energy drives emission across the entire electromagnetic spectrum, from radio wavelengths to the X-ray and γ-ray bands [Peterson, 1997, Frank et al., 2002]. Their bolometric luminosities span a wide range, from $\sim 10^{42}\,\mathrm{erg\,s^{-1}}$ for low-luminosity Seyfert nuclei to $\sim 10^{48}\,\mathrm{erg\,s^{-1}}$ for the most powerful quasars [Peterson, 1997, Netzer, 2015]. The standard unified model of AGN Antonucci [1993], Urry and Padovani [1995] posits that the central engine consists of a SMBH surrounded by an optically thick, geometrically thin accretion disc Shakura and Sunyaev [1973], itself surrounded by a dusty molecular torus. The observational diversity of AGN is largely attributed to the orientation of the line of sight with respect to the torus axis [Antonucci, 1993, Urry and Padovani, 1995]. Type 1 AGN (Seyfert 1, quasars) are viewed face-on, allowing a direct view of the broad-line region (BLR) and the continuum source, while Type 2 AGN are viewed edge-on, obscuring the inner regions [Antonucci, 1993].

1.2 X-ray Emission from AGN

The primary hard X-ray emission in (AGN) is mainly produced through inverse Compton scattering, where low-energy UV/soft X-ray photons from the accretion disc gain energy by interacting with hot electrons in a compact region called the corona Haardt and Maraschi [1991]. This process generates a power-law spectrum with photon index $\Gamma \approx 1.5$–2.5, extending to high energies set by the electron temperature kT_e. Typically, a high-energy cutoff is observed at $E_{\mathrm{cut}} \sim 2 - 3kT_e$ [Petrucci et al., 2001, Rybicki and Lightman, 1979].

Although inverse Compton scattering dominates the hard X-ray emission, other physical processes can also contribute. Thermal emission from the accretion disc provides the seed photons for Comptonization. In addition, weaker processes such as bremsstrahlung and syn-

chrotron radiation may occur in the corona, and synchrotron photons can also be up-scattered through synchrotron self-Compton processes Rybicki and Lightman [1979]. In very compact and luminous coronae, electron–positron pair production can become important and help regulate the temperature and optical depth of the plasma Lightman and Zdziarski [1987]. However, these processes generally play a secondary role compared to inverse Compton scattering.

The X-ray spectra of AGN also show several additional components beyond the primary power law. These include a soft X-ray excess below ∼2 keV, Compton reflection from the accretion disc producing a characteristic hump around 20–30 keV, and a fluorescent iron Kα emission line at ∼6.4 keV. When this reflection originates close to the black hole, relativistic effects broaden and skew these features, providing a powerful probe of the inner accretion flow and black hole spin Fabian et al. [1989].

1.3 Outflows in AGN

1.3.1 Warm Absorbers

Warm absorbers are clouds of photoionized gas within AGN that cause absorption at soft X-ray energies Halpern [1984]. They were first identified through absorption edges of O VII and O VIII in the soft X-ray spectra of Seyfert galaxies, and have since been found in approximately 50 per cent of Seyfert 1 galaxies observed with ASCA Reynolds [1997]. High-resolution spectroscopy with Chandra Low Energy Transmission Grating Spectrometer (LETGS) and XMM-Newton Reflection Grating Spectrometer (RGS) has resolved discrete narrow absorption lines in the soft X-ray spectra of several nearby AGN, enabling precise measurements of ionization parameters, column densities, and outflow velocities. Warm absorbers are characterized by their ionization parameter $\xi = L_{\rm ion}/(nr^2)$ Tarter et al. [1969], where $L_{\rm ion}$ is the ionizing luminosity, n is the gas density, and r is the distance from the ionizing source. Typical values span $\log \xi \approx 0$–3 erg cm s^{-1} [Blustin, 2004]. The absorbing gas is generally outflowing at velocities of a few hundred to a few thousand km s^{-1}, consistent with torus winds or accretion disc winds launched at parsec-scale distances from the black hole [Krolik and Kriss, 2001, Ashton et al., 2004]. However, in some cases, highly ionized absorption in the soft X-ray band also reveals ultra-fast outflows (UFOs), indicating the presence of multi-phase winds spanning a wide range of ionization states [Reeves et al., 2020, Pinto et al., 2018].

1.3.2 Ultrafast Outflows

Ultrafast outflows (UFOs) are highly ionized winds with velocities reaching a significant fraction of the speed of light. They are most commonly identified through blueshifted absorption features in the Fe K band, particularly transitions of Fe,XXV (Heα) and Fe,XXVI (Lyα) at rest-frame energies of approximately 6.70 and 6.97 keV, respectively [Tombesi et al., 2010b, Gofford et al., 2013]. They are defined as outflows with $v_{\rm out} > 0.1c$ [Tombesi et al., 2010b]. UFOs were first systematically catalogued by Tombesi et al. [2010b] in a survey of 42 local Seyfert galaxies observed with XMM-Newton, who found a detection rate of ∼ 35–40 per cent.

The outflow velocities of UFOs are determined from the relativistic Doppler formula. The observed absorption line energy $E_{\rm obs}$ is first converted to the AGN rest frame Tombesi et al. [2010b]:

$$E_{\rm rest} = (1 + z_{\rm source}) \times E_{\rm obs}, \tag{1.1}$$

where $z_{\rm source}$ is the cosmological redshift of the source. The blueshift of the absorber relative to the source is then:

$$1 + z_{\rm UFO} = \frac{E_{\rm ion}}{E_{\rm rest}}, \tag{1.2}$$

and the outflow velocity follows from the relativistic Doppler formula Tombesi et al. [2010b], Gofford et al. [2013]:

$$\frac{v_{\rm out}}{c} = \frac{(1 + z_{\rm UFO})^2 - 1}{(1 + z_{\rm UFO})^2 + 1}. \tag{1.3}$$

For warm absorbers detected via the `zxipcf` model, the intrinsic outflow velocity is obtained by correcting the absorber redshift $z_{\rm abs}$ for the source cosmological redshift:

$$1 + z_{\rm UFO} = \frac{1 + z_{\rm abs}}{1 + z_{\rm source}}. \tag{1.4}$$

1.3.3 Multi-phase Outflows

Several luminous AGN have been found to host multi-phase outflows in which the soft X-ray warm absorber and the highly ionized Fe K UFOs are co-spatial and co-moving, indicating a stratified wind spanning a wide range of ionization states [Tombesi et al., 2013, Gofford et al., 2013, Reeves et al., 2020, Pinto et al., 2018]. These winds are interpreted as large-scale accretion disc outflows in which different ionization zones are observed simultaneously along the line of sight. A prototypical example is the quasar PDS 456 ($z = 0.184$, $L_{\rm bol} \sim 10^{47}$ erg s^{-1}), where high-resolution X-ray spectroscopy has revealed multiple absorption features associated with both soft X-ray ions and Fe K transitions [Reeves et al., 2009, Matzeu et al., 2016, Reeves et al., 2020]. In particular, blueshifted absorption lines of O VIII Lyα, Ne IX Heα, and Ne X Lyα detected in the XMM-Newton/RGS spectrum trace a fast, highly ionized wind with velocity $v_{\rm out}/c \approx -0.25$, consistent with the velocity inferred from Fe XXV and Fe XXVI absorption in the hard X-ray band [Matzeu et al., 2016, Reeves et al., 2020]. The agreement in velocity between the soft X-ray and Fe K absorbers strongly supports a multi-phase, co-moving outflow originating from a common physical structure. Such winds are consistent with wide-angle accretion disc outflows launched at sub-parsec scales, where different ionization components arise due to density and illumination gradients within the flow [Tombesi et al., 2010b, Fukumura et al., 2010].

1.4 AGN Feedback

The presence of multi-phase, co-moving outflows in AGN suggests that these winds are not merely a local phenomenon confined to the vicinity of the black hole, but can have far-reaching consequences on galactic scales. In particular, such stratified winds provide a natural channel through which energy and momentum released during accretion can be transported outward,

linking the small-scale physics of the accretion disc to the large-scale evolution of the host galaxy [Tombesi et al., 2013, Fukumura et al., 2010, Reeves et al., 2020]. This connection is supported by observational evidence for the co-evolution of SMBHs and their host galaxies, most notably through the M–σ relation [Ferrarese and Merritt, 2000, Gebhardt et al., 2000], which indicates a tight correlation between the black hole mass and the stellar velocity dispersion of the galactic bulge. This relation implies that the growth of the SMBH and the host galaxy are regulated through a common physical mechanism [Kormendy and Ho, 2013, Heckman and Best, 2014]. UFOs, as part of these multi-phase winds, are strong candidates for driving such feedback [Tombesi et al., 2013, King and Pounds, 2015]. Owing to their high velocities and large ionization states, UFOs carry substantial kinetic energy that can couple to the surrounding interstellar medium [Zubovas and King, 2012, Faucher-Giguère and Quataert, 2012]. As these winds propagate outward, they can shock and accelerate ambient gas, giving rise to large-scale molecular outflows observed on kiloparsec scales [**?**Tombesi et al., 2015, Faucher-Giguère and Quataert, 2012]. This process can deplete or heat the cold gas reservoir required for star formation, thereby suppressing or quenching star formation activity in the host galaxy [Tombesi et al., 2015, Zubovas and King, 2012]. For AGN feedback to be effective, the kinetic power of the outflow must constitute a significant fraction of the AGN bolometric luminosity. Both theoretical models and observational studies suggest that a threshold of approximately ~ 0.5–5 per cent of the bolometric luminosity is required for the outflow to efficiently couple with the galactic gas and drive galaxy-scale feedback [Hopkins and Elvis, 2010]. When this condition is satisfied, AGN-driven winds can play a central role in regulating galaxy evolution, establishing a direct physical link between black hole accretion and the properties of the host galaxy.

1.5 Scientific objectives and outline of the thesis

UFOs are increasingly detected in a large number of AGN. Also, scaling relations are now established between various outflow properties and host galaxy properties. These scaling relationship are based on about 100 AGN and therefore it is important to increase the number of AGN with known UFOs. The thesis therefore aims to 1. Increase the number of AGN with known UFOs: For this I have carried out a systematic investigation of the X-ray spectra of a larger number of sources, with the prime motivation of finding the presence of outflows in such systems. My initial sample consists of 182 AGN with data in the XMM-archive with more than 50 ksec. of observations. From phenomenological fits, I found possible evidence of UFOs in 26 sources. Physical model fits were carried out on these 26 sources to refine the various physical models that need to be used. Initial analysis has led to the first time detection of UFO is a luminous quasars. The results on this source is detailed in Chapter 5 of the thesis.

Chapter 2

Catalogue Used, Sample Selection and Data Reduction

2.1 Catalogue Used

We started with the million quasars catalogue. The Million Quasars (Milliquas) Catalog is one of the most comprehensive compilations of known and candidate AGN, designed to provide a unified and continuously updated resource for AGN studies Flesch [2023]. It combines data from a wide range of large-scale multiwavelength surveys, including optical surveys such as the Sloan Digital Sky Survey (SDSS; York et al. [2000]), as well as radio and X-ray observations. The catalog includes spectroscopically confirmed quasars along with high-confidence photometric candidates, each assigned a probability of being a quasar, enabling statistically robust sample selection. With over a million entries, the Milliquas catalog is particularly valuable for cross-matching with high-energy observatories such as *XMM-Newton, NuSTAR*, facilitating studies of accretion physics, obscuration, and UFOs in AGN. Its broad sky coverage and integration of heterogeneous datasets makes it an essential tool for both targeted investigations and large-sample statistical analyses in extragalactic astrophysics.

2.2 Sample Selection

Firstly we started with the Milliquas catalogue which has a total of 10,21,800 AGN. Since, In this thesis work, our goal is to detect, do outflows studies and see how outflows affect their own galaxies. To do this, we need the the data from XMM-Newton achieve becuase the outflows signature appears in the range of 6-10 keV which is well covered by XMM-Newton. Therefore, we do need the data of XMM-Newton. Now, to get the source, we cross-correlated the milli-quasars catalogue with XMM-Newton so that we gets the entry which reported by the XMM-Newton. We performed a systematic positional cross-match against the XMM-Newton Master Catalog (xmmmaster) hosted by the High Energy Astrophysics Sci- ence Archive and Research Center (HEASARC) at NASA. This automated pipeline queries the HEASARC database for each candidate source coordi- nate and retrieves all XMM-Newton detections within a fixed

angular separation threshold, producing a complete record of matches, non-detections, and query failures.To cross-correlate the milli-quasats catalogue with XMM-Newton we wrote the code and provide the search radius of 5" for sources to get find.
Once the code is done then we run the code and it gave us the 911 entries which were detected in XMM-Newton.

These 911 entries has the unique AGN recoded in XMM-Newton but at this point of time, we don't know actually the data is present or not or XMM-Newton just pointed the source and did not collect the data. Therefore, we need to apply the condition on exposure time so that we can get the sufficient amount of counts for UFO detection. So, we applied two condition: (1). data should be achieved, and (2) the duration must be $\geq$ 40 ks. Through this, we finally landed with 473 unique sources.

Now these 473 are the sources which are points of interests. Now, we have downloaded the data for all 473 source and reduced the data of detector EPIC-pn and we finally analysed that among 473 sources, there are sources which are not have sufficient counts in 6-10 keV band which is the region of UFO detection, low luminiouity sources. Therefore, after analysis this pattern, we ended up with 182 sources. Now, we checked the each sources with their obsids individually with model:

$$\text{tbabs} \times \text{ztbabs} \times \text{powerlaw} \tag{2.1}$$

where `tbabs` represents the Galactic photoelectric absorption along the line of sight,
`ztbabs` accounts for the intrinsic absorption at the redshift of the source, and
`powerlaw` describes the primary X-ray continuum emission, typically arising from inverse Compton scattering of disk photons in a hot corona. I used the HI4PI suvery for galaxtic absorption (nH).

After analysising the sources with the model present above, we see the spectra which is diverted from the powerlaw and showing the signature of furture analysis with advanced model. One of the most important part here is that as we know that the UFO absoprton line lies in 7 to 10 keV and this is also the region of reflection....**put more here and citation** . And this can artifact the absortion line since reflection and absorption both fights in this region, therefore, we applied the model:

$$\text{tbabs} \times \text{ztbabs} \times (\text{nthcomp} + \text{xillverCp / relxillCp}) \tag{2.2}$$

The details of the individual model is present in Table 2.1. Then we ended up with 30 sources that can actually has the outflows and are survived. Now there is 90% chances that these sources have the outflows, we don't know may be normal velocity outflow or ultrafast outflows.

Therefore, We need to find the sources that has the outflows and has not been published yet, since many sources has more than one obsids and there are chances that the outflow may occurred in one obsid and may not be occurred in other obsids, this may be due to the flux change/variablity or high brightness in that time and all line may completely ionized.

After doing the literature review of these 30 sources, we ended up with 26 sources which

Table 2.1: Spectral models used in this work and their key properties.

heightModel	Description	Key Parameters
`nthcomp`	Thermal Comptonization continuum produced by up-scattering of seed photons by hot coronal electrons.	Γ, kT_e, $kT_{\rm seed}$
`xillverCp`	Ionized reflection spectrum from an accretion disc illuminated by a Comptonized continuum.	$\log \xi$, $A_{\rm Fe}$, Γ, kT_e
`relxillCp`	Relativistic reflection model including Doppler shifts, gravitational redshift, and light bending effects near the black hole.	a, i, $R_{\rm in}$, $\log \xi$, Γ, kT_e

are highly prominent to constains the outflows and not yet published. The list of sources is presented in Table **??**.

For search radius: We need to give a correct value of search radius so that the other sources don't come in the field of view of a source. In order to do this, we need to use three terms:

2.2.1 Astrometric Accuracy:

The positional uncertainty of a source is primarily determined by the astrometric accuracy of the instrument. For *XMM-Newton*, this is typically ~ 1–$2''$, which sets the baseline uncertainty in source localization.

2.2.2 Point Spread Function (PSF):

The finite PSF of the telescope broadens the source image, introducing additional uncertainty in centroid determination. For *XMM-Newton*, the PSF has a typical FWHM of $\sim 6''$, contributing to positional dispersion.

2.2.3 Source Density / Confusion:

The probability of false associations increases with source density in the field, especially in crowded regions. A smaller search radius helps minimize chance coincidences while retaining true counterparts.

The combined positional uncertainty can be approximated as:

$$r \approx \sqrt{\sigma_{\rm astro}^2 + \sigma_{\rm PSF}^2} \tag{2.3}$$

For *XMM-Newton*, taking $\sigma_{\rm astro} \sim 2''$ and $\sigma_{\rm PSF} \sim 4''$ (centroid-level), we obtain:

$$r \approx \sqrt{(2'')^2 + (4'')^2} \approx 4.5'' \tag{2.4}$$

A search radius of $5''$ is adopted as it encompasses the combined astrometric and PSF-driven positional uncertainties of *XMM-Newton*. a matching radius of 5 arcsec therefore encompasses

more than 3σ of the combined positional uncertainty of both the input catalog and the XMM source, ensuring that physically associated counterparts are not rejected due to astrometric scatter.

Now, after putting this RA and DEC. We wrote the code which cross-correlated with XMM-Newton at search radius of 5".

2.2.4 Details of Code:

Code was written with high-precision so that we don't get any problem in source or it dont give any false/fake sources or it dont remove the source from the catalogue. we performed a systematic positional cross-match against the XMM-Newton Master Catalog (xmmmaster) hosted by the High Energy Astrophysics Sci- ence Archive and Research Center (HEASARC) at NASA. This automated pipeline queries the HEASARC database for each candidate source coordi- nate and retrieves all XMM-Newton detections within a fixed angular sep- aration threshold, producing a complete record of matches, non-detections, and query failures.

Catalog Description

The `xmmmaster` catalog is the master log of the XMM-Newton Slew Survey compiled from observations taken by the European Space Agency's XMM-Newton space observatory between 2000 and the present. It contains approximately 520,000 X-ray point sources detected by the EPIC (European Photon Imaging Camera) instrument suite—comprising the EPIC-pn, EPIC-MOS1, and EPIC-MOS2 detectors—operating in the 0.2–12 keV energy band. All source positions in this catalog have been astrometrically refined through multi-epoch data and carry a systematic positional uncertainty of approximately 1.5–2.0 arcseconds, consistent with the instrument's on-axis point spread function (PSF) full-width at half-maximum (FWHM) of ≈ 6 arcseconds.

The catalog is accessed programmatically through the `astroquery.heasarc` Python interface, which submits Virtual Observatory cone-search queries to the HEASARC web service and returns machine-readable tabular results.

Methodology

The cross-matching pipeline executes the following sequence of operations for each candidate source:

1. **Input loading and validation.** The pipeline reads a CSV file containing `RA` and `DEC` columns in decimal degrees.

2. **Checkpoint-based resumption.** Previously processed indices are loaded from an existing summary file and excluded from the current run, allowing interrupted sessions to resume without duplicating completed queries.

3. **Coordinate standardisation.** Each coordinate pair is converted into an `astropy.coordinates.SkyCoor` object:

   ```
   coord = SkyCoord(ra=ra, dec=dec, unit=("deg", "deg"))
   ```

   This anchors the position to the ICRS/J2000.0 frame used by the XMM-Newton catalog and validates that RA $\in [0°, 360°)$ and DEC $\in [-90°, +90°]$.

4. **HEASARC cone-search query.** A cone-search is submitted to the HEASARC web service:

   ```
   result = heasarc.query_region(coord, catalog="xmmmaster",
                                 radius="5 arcsec")
   ```

   All `xmmmaster` sources within 5 arcseconds are returned, with separations computed via the exact great-circle distance:

   $$d = \arccos(\sin\delta_1 \sin\delta_2 + \cos\delta_1 \cos\delta_2 \cos\Delta\alpha)\,. \tag{2.5}$$

5. **Result parsing and status assignment.** Successful queries with at least one match are tagged `SUCCESS_WITH_MATCHES`; those returning no sources are tagged `SUCCESS_NO_MATCHES`. Exceptions are categorised by failure type (Section 2.2.4).

6. **Rate-limiting.** A 100 ms delay between consecutive queries prevents exceeding the HEASARC request rate and avoids HTTP 429 errors that would otherwise appear as false non-detections.

7. **Parallel processing and checkpointing.** Queries run across multiple worker processes via Python's `multiprocessing` module. Results are written to disk every N entries (default $N = 100$), bounding data loss in the event of failure.

Note: I also cross-verify the catalogue after the cross-check and found no two sources have the same ra dec, And also I used SIMBAD and NED for cross-check the names of source by putting RA DEC since people do the multi-observation, and based on the satellite used for data taking, people often name the source according, that's why I cross-check the source name and RA DEC and most importantly redshift of source.

Reliability and Accuracy of Results

- **Astrometric standardisation.** All coordinates are converted to `SkyCoord` objects anchored to the ICRS/J2000.0 frame before querying, ensuring consistency with the XMM-Newton catalog reference frame and preventing systematic positional offsets.

- **Complete match retrieval.** The query returns *all* sources within the 5 arcsecond aperture, not only the nearest. Cases with two or more matches are flagged via the `matches_found` field for manual inspection and disambiguation using source flux, hardness ratio, or spatial extent.
- **Unambiguous status codes.** The pipeline explicitly distinguishes `SUCCESS_NO_MATCHES` (query completed; no source found) from `SERVER_ISSUE` (query failed; outcome unknown). This prevents network failures from being misclassified as confirmed non-detections.

Error Classification and False-Result Prevention

All exceptions raised during query execution are caught and classified into one of six categories based on keywords in the exception message. Network-related failures such as timeouts, SSL errors, and HTTP 502/503/504 responses are recorded as `SERVER_ISSUE` and require the query to be retried after a delay. Malformed query strings are flagged as `SYNTAX_ERROR`, while unrecognised or out-of-range coordinate values are assigned `COORDINATE_ERROR`, both of which require inspection and correction of the input data. Cases where the returned result set exceeds available system memory are marked `MEMORY_ERROR` and can be resolved by reducing the processing batch size. API quota violations or HTTP 403 responses are recorded as `ACCESS_ERROR`, indicating that the request rate should be throttled or the workload split across sessions. All remaining uncategorised exceptions are assigned `OTHER_ERROR` for manual review.

Failed entries are written to a dedicated error CSV containing only the columns `index`, `RA`, `DEC`, and `error_message`, which can be passed directly as input to a subsequent pipeline run. This ensures that every entry in the final `SUCCESS_NO_MATCHES` set represents a confirmed non-detection, not a silently failed query

Limitations

Two physical limitations of the cross-matching procedure should be noted when interpreting the results. First, the `xmmmaster` catalog covers approximately 75% of the sky ($\approx 31{,}000\,\mathrm{deg}^2$), with gaps concentrated along the Galactic plane where X-ray absorption and source confusion are severe; sources in uncovered regions will always return `SUCCESS_NO_MATCHES` regardless of their true X-ray flux, so non-detections should be cross-referenced against the XMM survey footprint before being interpreted as upper limits. Second, the XMM-Newton Slew Survey reaches a point-source flux limit of approximately $10^{-11}\,\mathrm{erg\,s^{-1}\,cm^{-2}}$ in the 0.2–12 keV band; sources fainter than this threshold are undetectable irrespective of their intrinsic X-ray emission, and their absence from the match results reflects an instrumental sensitivity limit rather than a physical non-detection. Some queries that initially failed due to network or server errors were reprocessed in a subsequent run using the dedicated error CSV as input, ensuring that the final catalog is complete and that every source has received a confirmed query outcome.

2.3 XMM-Newton And Data Reduction

2.3.1 XMM-Newton

XMM-Newton [X-ray Multi-Mirror Mission; , Jansen2001] is an ESA cornerstone mission launched on 10 December 1999 into a highly elliptical orbit with a period of approximately 48 hours and an apogee of ∼114,000 km, designed to allow long, uninterrupted observations above the Earth's radiation belts. The spacecraft carries three co-aligned X-ray telescopes, each consisting of 58 nested Wolter type-I mirror shells with a combined on-axis effective area of approximately 4500 cm^2 at 1 keV and ∼1800 cm^2 at 6 keV, making it the largest collecting area X-ray observatory ever flown at the time of its launch. The focal plane instruments include the European Photon Imaging Camera [EPIC; Strüder et al., 2001, Turner et al., 2001], which comprises one EPIC-pn camera and two EPIC-MOS cameras operating simultaneously over the 0.2–15 keV energy band with a field of view of ∼30 arcmin diameter, angular resolution of ∼6 arcsec FWHM and 15 arcsec half-energy width, and a timing resolution of 0.03 ms for EPIC-pn in timing mode. The EPIC-pn camera Strüder et al. [2001] uses a monolithic silicon p-n junction CCD array of 12 chips and provides the highest count rate capability among the EPIC instruments, with an effective area of ∼1400 cm^2 at 1 keV and ∼800 cm^2 at 6 keV, an energy resolution of $\Delta E \approx 70$ eV at 1 keV and ∼150 eV at 6 keV, and a sensitivity to point sources as faint as $\sim 5 \times 10^{-15}$ erg cm^{-2} s^{-1} in the 0.5–2 keV band for a 100 ks observation Strüder et al. [2001]. This combination of large effective area and moderate spectral resolution makes EPIC-pn the primary instrument for the detection of broad absorption features in the Fe K band, such as the blueshifted Fe XXVI Lyα lines associated with ultrafast outflows at $E_{\rm rest} \sim 7$–10 keV Tombesi et al. [2010a], Gofford et al. [2013], Nardini et al. [2015], and for the characterization of the broadband X-ray continuum from the corona and accretion disc. The EPIC-MOS cameras Turner et al. [2001] each provide a complementary effective area of ∼500 cm^2 at 1 keV with a slightly superior energy resolution of $\Delta E \approx 50$ eV at 1 keV and a point source sensitivity comparable to EPIC-pn Turner et al. [2001], and are used for cross-calibration verification and joint broadband spectral fitting Piconcelli et al. [2005]. Two of the three X-ray telescopes are also equipped with Reflection Grating Spectrometers [RGS; den Herder et al., 2001], which intercept approximately 40 per cent of the incoming X-ray flux before it reaches the MOS cameras and disperse it onto two dedicated back-illuminated CCD strip detectors, providing high-resolution soft X-ray spectroscopy over the wavelength range 6–38 Å (0.33–2.5 keV). The RGS achieves a resolving power of $\lambda/\Delta\lambda \approx 100$–500 (corresponding to a velocity resolution of ∼600–3000 km s^{-1}), which is sufficient to resolve individual narrow absorption and emission lines from warm absorbers den Herder et al. [2001], Blustin [2004], including the O VII Heα triplet at 21.6 Å, O VIII Lyα at 18.97 Å, Ne IX Heα at 13.45 Å, and Ne X Lyα at 12.13 Åthat are the primary diagnostics of photoionized warm absorbers in AGN Kaastra et al. [2000], den Herder et al. [2001]. The effective area of the combined RGS1 and RGS2 detectors peaks at ∼170 cm^2 near 15 Å and falls off steeply towards shorter wavelengths, reaching ∼30 cm^2 at 6 Å den Herder et al. [2001]. The RGS sensitivity to warm absorber features depends critically on the source count rate Blustin [2004] — for sources with EPIC-pn count rates above ∼0.5 counts s^{-1}, the RGS

can detect ionic column densities as low as $N_{\rm ion} \sim 10^{15}$–10^{16} cm^{-2} in a 50 ks observation. The spacecraft also carries an Optical Monitor [OM; Mason et al., 2001], a 30 cm Ritchey-Chrétien UV/optical telescope providing simultaneous photometric and grism spectroscopic coverage from 170 to 650 nm in six broadband filters Mason et al. [2001], enabling multi-wavelength characterization of the AGN spectral energy distribution. XMM-Newton is operated by the European Space Astronomy Centre (ESAC) in Villanueva de la Cañada, Spain, with observations selected through competitive peer review and archived in the XMM-Newton Science Archive (XSA). In the present study, XMM-Newton EPIC-pn data are used as the primary instrument for the detection and characterization of ultrafast outflows through blind Fe K line searches in the 6–10 keV rest-frame energy band Tombesi et al. [2010b], Gofford et al. [2013], for the measurement of the broadband X-ray continuum parameters (Γ, kT_e, reflection fraction), and for the spectral modelling of warm absorbers in the soft X-ray band below 2 keV Blustin [2004], Piconcelli et al. [2005]. For the blazar H2356−309, joint EPIC-pn and RGS spectroscopy is employed to simultaneously constrain the warm absorber column density, ionization parameter, and outflow velocity through the resolved soft X-ray absorption lines. The large effective area of EPIC-pn at 6–7 keV is essential for the detection of the faint (EW $\sim$ 30–100 eV) blueshifted Fe K absorption features that are the hallmark of UFOs Tombesi et al. [2010b], Nardini et al. [2015], while the high spectral resolution of the RGS provides complementary constraints on the low-ionization warm absorber phases that are below the spectral resolution threshold of EPIC-pn den Herder et al. [2001], Blustin [2004]. All data were processed using the Science Analysis System [`SAS`; Gabriel et al., 2004] version 22.1.0 following the standard pipeline procedures described in Chapter 2.

2.3.2 EPIC-pn Data Reduction

Firstly for to work with XMM-Newton, you need to setup the SAS interface with Heasoft tools. Once this is done, now you need to download the calibration from Calibration Files. Now you need to compile these in your system and add this in bashrc.

Data reduction of EPIC-PN: Initialize the heasoft. Then set SAS directory to your actual directory where you are working.

Step 1: Create Working Directory and Setup

Initialize the heasoft and set the directory where you are working.

Step 2: Generate Calibration Index

when you download the ODF file from XMM-Newton then extract it and it will give some files with another compressed files. Now again extract this new compressed file and make new folder and put the file you got from this new extraction and from the old extraction.
Command:

```
# Generate calibration index file
cifbuild
```

The `cifbuild` task in the XMM-Newton SAS software is used to generate the Calibration Index File (CCF), which tells SAS which calibration files are required for a given observation. When `cifbuild` is run, it scans the observation data directory (specified by `SAS_ODF`) and the calibration directory (specified by `SAS_CCFPATH`) to determine the appropriate calibration files that correspond to the instrument configuration and observation date. It then compiles this information into a single file named `ccf.cif`. This file acts as a lookup table that SAS uses to correctly calibrate the event files during data reduction tasks such as `rgsproc` or `epproc`. Without `cif.cif`, SAS would not know which calibration products to apply, and the data could not be processed accurately.
Now set the new generated calibration file to main calibration file using the command:

```
export SAS_CCF=~/Desktop/quasars_thesis/sources/1H0707-495/0148010301_obsid/ccf.cif
```

Step 3: Generate Observation Summary

For to generate the observation files, we would use the command:

```
# Create summary file
odfingest
```

The `odfingest` task processes the raw Observation Data Files (ODF) provided by XMM–Newton and builds a summary file that is required for all subsequent SAS data reduction steps. It reads the instrument housekeeping information, camera configurations, and timing metadata contained in the original ODF files and compiles them into a single Observation Summary File (the file ending in `*SUM.SAS`). This summary file provides SAS with a structured index of all exposures and calibration information associated with the observation. After running `odfingest`, the environment variable `SAS_ODF` must be updated to point to the newly created `*SUM.SAS` file so that tools like `epproc` and `rgsproc` can correctly reference the observation setup. In short, `odfingest` standardizes and organizes the raw ODF directory into a format that SAS tasks can understand and process. Now set the new generated observation file to main observation file using the command:

```
# This will create something like: 1387_0506200401_SCX00000SUM.SAS
export SAS_ODF=~~/Desktop/quasars_thesis/sources/1H0707-495/0148010301_obsid/1387
    _0506200401_SCX00000SUM.SAS
```

Output: Should show something like: `*_0148010301_SCX00000SUM.SAS`

Step 4: Generation of calibrated event files

The XMM-Newton data were processed using the Science Analysis System (SAS). The Observation Data Files (ODF) were first ingested using the `odfingest` task to generate the summary file required for further processing. The EPIC-pn data were reduced using the `epproc` pipeline, while the MOS1 and MOS2 data were processed using `emproc`, producing calibrated event lists for each instrument. The Reflection Grating Spectrometer (RGS) data were processed using the `rgsproc` task, which generates calibrated spectra along with the corresponding response and background files. These processed event files were then used for subsequent spectral analysis. **Since, For my thesis work, I am using the epic-pn detector's only, therefore, I will go through the next steps for epic-pn only.**

Step 5: Process EPIC-pn Data

To generate the calibrate event file for epic-pn data, we ahve to use the command:

```
epproc
```

The `epproc` task in the XMM-Newton SAS pipeline is used to process the raw Observation Data Files (ODF) from the EPIC-pn detector and convert them into calibrated event lists that are ready for scientific analysis. When `epproc` is run, it applies all necessary instrument calibrations, time and energy corrections, removes bad pixels, flags bad events, and determines the position and energy of each detected X-ray photon. It also generates associated housekeeping information such as attitude data and good time intervals. The output of `epproc` is a cleaned and calibrated EPIC-pn event file, which can then be used for spectrum extraction, light curve creation, and imaging. In short, `epproc` performs the essential conversion from raw detector readout to scientifically usable event data.

Output: `P0148010301PN000EVLI0000.FIT`, `*EPN*ImagingEvts.ds`

Step 6: Check for Background Flaring

XMM-Newton has the coverage from 0.3 15 keV but after the 10 keV, there are flares which are noise, therefore we need to see the flare so that we can remove high flare and our data dont get contaminated. The command to do this is:

```
# Create background light curve
evselect table=$(ls *EPN*ImagingEvts.ds) \
withrateset=yes rateset=pn_ltcrv_bkg.fits \
maketimecolumn=yes timebinsize=100 makeratecolumn=yes \
expression='#XMMEA_EP && (PI>10000&&PI<12000) && (PATTERN==0)'
```

`evselect` is command. **Input:** `*EPN*ImagingEvts.ds` – Input event file (EPIC-pn cleaned event list), `withrateset=yes` – Tell SAS to produce a rate (light curve) file, **Output:** `pn_ltcrv_bkg.fits` – Name of the light curve output file (This is the background light curve file you will use to identify flares), `expression='#XMMEA_EP && (PI>10000&&PI<12000) && (PATTERN==0)'`: Event

selection filter – Removes bad pixels/events and selects high-energy single-pixel events to trace background flares.

Understanding the Filter Expression

- `#XMMEA_EP`: Removes bad/hot pixels and noise events
- `PI > 10000 && PI < 12000`: Selects events in 10–12 keV energy range (At these high energies, the source contributes almost nothing → this measures the background only)
- `PATTERN == 0`: Selects single-pixel events only (Gives the cleanest, most reliable flare indicator)

Why This Step Is Important

The background in EPIC-pn often rises due to solar particle flares. If we do not remove these intervals → the final spectrum becomes WRONG. We use the 10–12 keV background-only light curve to determine which time intervals to keep and which to throw away.

Step 7: Plot Light Curve

Here we do use the dsplot to see the light curve fits.

```
dsplot table=pn_ltcrv_bkg.fits x=TIME y=RATE
```

To visually inspect the flare activity. Note: Stable background: Flat or small fluctuations (Data is clean → continue) and Soft-proton flare periods: Sharp spikes or high count rate periods (These must be removed using GTI filtering). For good, people use the threshold < 0.4 cts/s. If anything above you see then that must be removed by GTI filtering.

Step 8: Create GTI File

Based on what you see in the plot, create GTI:

```
# Standard threshold (adjust if needed based on plot)
tabgtigen table=pn_ltcrv_bkg.fits expression='RATE<=0.4' gtiset=pn_gti.fits
```

Input: `pn_ltcrv_bkg.fits` (created in background flare), **Output:** `pn_gti.fits` (this is still background flare fits file but with no flares), `RATE<=0.4`: remove all flare above it.

GTI used to remove the flares. Now open output file in fv and see the exposure time. Fv is fit viewer, an important to see the exposure because it will tell that how much exposure we have lost. To do this, open fv and select gtiset file, then click on extension "GTI" then Table and there will be START and STOP icon and you will see the exposure (imp. is there will be interval of exposure because after removal of flare interval we see).

Step 9: Create Filtered EPIC-pn Event File

This is very important as it will make the final file for XSPEC analysis. Command is mentioned below;

```
evselect table=$(ls *EPN*ImagingEvts.ds) \
withfilteredset=yes filteredset=pn_clean.fits \
destruct=yes keepfilteroutput=yes \
expression='#XMMEA_EP && (PATTERN<=4) && (FLAG==0) && (PI>150) && gti(pn_gti.fits,TIME
    )'
```

`(PATTERN<=4)`: Keeps single and double pixel events only (These are high-quality X-ray photon events), `(PI>150)`: Removes events below ∼150 eV (Prevents soft electronic noise contamination), `gti(pn_gti.fits,TIME)`: Apply Good Time Intervals derived from flare screening (Removes high-background flare periods).

The EPIC-pn event file was then filtered to remove periods of high background and to retain only high-quality X-ray events suitable for scientific analysis. This was done using the `evselect` task to apply both standard event screening criteria and the Good Time Intervals (GTIs) derived from the background light curve. Specifically, we selected only events that satisfy the recommended EPIC-pn quality flag (`#XMMEA_EP`), restricted the event patterns to single and double pixel events (`PATTERN` ≤ 4), and excluded any events flagged as unreliable (`FLAG == 0`). To avoid low-energy detector noise, only events with energies above 150 eV were retained (`PI > 150`). The GTI file (`pn_gti.fits`) was included to remove time intervals affected by soft-proton flares. The result of this filtering procedure is the cleaned and calibrated EPIC-pn event file (`pn_clean.fits`), which forms the basis for all subsequent spectral and timing analysis.

Step 10: Check Resulting Exposure

```
fkeyprint pn_clean.fits+1 EXPOSURE
fkeyprint pn_clean.fits[1] EXPOSURE
fkeyprint pn_clean.fits[1] TSTART
fkeyprint pn_clean.fits[1] TSTOP

\section{Step 10: Create Image for Source Selection}
COmmand we do use:
\begin{lstlisting}[language=bash]
evselect table=pn_clean.fits imageset=pn_image.fits \
xcolumn=X ycolumn=Y imagebinning=binSize \
ximagebinsize=80 yimagebinsize=80
```

The cleaned EPIC-pn event file was used to generate an X-ray image for the purpose of defining accurate source and background extraction regions. This was accomplished using the `evselect` task to bin the detected photon positions in detector coordinates (columns X and Y) into an image file (`pn_image.fits`). A binning scale of 80 detector pixels per image bin was chosen intentionally — it reduces pixel-to-pixel statistical noise and smooths the image

sufficiently to clearly reveal the shape and centroid of the source emission, while still preserving the spatial structure needed for precise region selection. This step is important because the reliability of the extracted spectrum depends critically on choosing a source region that captures all genuine source photons and a background region that is free from contamination by the source or nearby objects. The resulting image was then examined in DS9 to interactively select a circular source region centered on the peak brightness and a nearby background region on the same CCD to ensure accurate background subtraction.

Step 11: Open in DS9

```
ds9 pn_image.fits &
```

In DS9, you will see only complete bright source in linear scale but sometimes source is faint so we need to use log scale (it maximize the fainter objects, this also maximize the brightness of fainter background). Open DS9 then choose scale to log then zoom according to source can be visible then go to file then select the region. Now point on the center of source and double click then a new window will open choose physical coordinate there (center of circle coordinate will come automatically and radius you put in arc sec then choose physical coordinate then DS9 will automatically convert it into physical radius) then save, again apply same method but now circle will be away from source in same CCD with larger radius then source's circle radius. Note down the coordinate if you want.
Be careful while doing this as your input ra dec should match with the source seen in the ds9, you can do this by just clicking on source and note down the ra dec and compare with your input ra dec.

Since the source lies near to the end of CCD, so it recommended to verify there is no artefacts arisen during source extarction. To do this, we use a tool names as "epatplot", see more below.

Step 12: Identification of PSF Artefacts in EPIC-pn Images

`epatplot` performs a pile-up analysis for EPIC-pn data. This is particularly useful to investigate whether the spike-like structures seen around bright sources in DS9 images represent real astrophysical jets or mirror-induced PSF artefacts from *XMM-Newton*.

```
epatplot set=pn_clean.fits plotfile=epat.ps device=/xs
```

The linear spike-like features seen around some bright sources in EPIC-pn images are not real structures from the AGN, but are caused by the telescope itself. These arise due to the mirror support structure of *XMM-Newton* and are commonly known as diffraction spikes or PSF spokes. We confirm their instrumental origin by checking that the `epatplot` results show the single and double event fractions are consistent with the model (within about $\pm 5\%$) in the 0.5–10 keV range, indicating that pile-up is not significant. In addition, these spikes are usually symmetric and aligned radially with respect to the detector centre, which matches the expected EPIC-pn PSF pattern. Similar features are also seen around other bright sources in the same

field, further confirming that they are not intrinsic to the source. However, care is taken in cases where the spikes appear asymmetric, unusually extended, or occur near CCD gaps or detector edges, as such effects can sometimes mimic real extended emission or require adjustment of the source extraction region.

A practical rule of thumb is therefore:

If `epatplot` shows no pile-up and the spokes are symmetric: the feature is instrumental.

If either condition fails: further investigation is needed.

Step 13: Extract EPIC-pn Source Spectrum and Background using coordinates

Now, we need to extract the region of source and background. Source region is extract with 30arcsec radius and background is extract with 50 arcsec. Background was extracted away from source so that the background does'nt contaminate the source. Background was extracted on the same CCD of source.

Command to use for source extarction:

```
evselect table=pn_clean.fits withspectrumset=yes \
spectrumset=pn_source_spec.fits energycolumn=PI spectralbinsize=5 \
withspecranges=yes specchannelmin=0 specchannelmax=20479 \
expression='(FLAG==0) && (PATTERN<=4) && ((X,Y) IN circle(24881.384,24062.745,600))'
```

Command to sue for background extraction:

```
evselect table=pn_clean.fits withspectrumset=yes \
spectrumset=pn_bkg_spec.fits energycolumn=PI spectralbinsize=5 \
withspecranges=yes specchannelmin=0 specchannelmax=20479 \
expression='(FLAG==0) && (PATTERN<=4) && ((X,Y) IN circle(22523.702,31430.5,1000))'
```

Input: `pn_clean.fits`, **Output:** `pn_source_spec.fits`, `pn_bkg_spec.fitsenergycolumn=PI`, `spectralbinsize=5`: Bin the energy channels in steps of 5 (Controls spectral resolution; balances S/N vs detail), `specchannelmin=0 specchannelmax=20479`: Use full EPIC-pn calibrated energy range.

The source spectrum was extracted from the cleaned EPIC-pn event file using the `evselect` task. A circular source region was defined based on the pn image, with the coordinates taken directly from the DS9 region file. Only high-quality X-ray events were selected using the criteria `FLAG==0` to exclude detector edge and bad pixel events, and `PATTERN<=4` to retain single- and double-pixel photon detections, which offer the best energy calibration. The photon energies were accumulated using the PI channel, with a spectral binning of 5 PI channels to balance spectral resolution and signal-to-noise. The resulting source spectrum was saved as `pn_source_spec.fits` and is used for subsequent spectral analysis.

Now, we need to generate the response matrix and ancillary response files.

Step 14: Generate RMF (Response Matrix)

After extracting the source and background regions (for example using DS9), the next important step is to generate the response files required for spectral analysis. One of these is the Redistribution Matrix File (RMF). The RMF describes how the detector records incoming X-ray photons, i.e., it gives the probability that a photon of a given true energy is detected in a particular energy channel. This is necessary because detectors do not measure the exact photon energy perfectly; instead, the energy is slightly redistributed due to instrumental effects. Therefore, without the RMF, the observed spectrum cannot be correctly compared with physical models. In XMM-Newton analysis, the RMF is generated using the `rmfgen` task, and it is used together with the Ancillary Response File (ARF) during spectral fitting to accurately model the source emission.

Create rmf file using the command:

```
rmfgen rmfset=pn_source.rmf spectrumset=pn_source_spec.fits
```

Input: `pn_source_spec.fits`, **Output:** `pn_source.rmf`

Source spectrum file is used to create the rmf file.

Step 15: Generate ARF (Ancillary Response)

After generating the RMF, the next step is to create the Ancillary Response File (ARF), which is also essential for spectral analysis. The ARF describes how efficiently the telescope and detector collect X-ray photons as a function of energy. In simple terms, it accounts for the effective area of the instrument, including factors such as mirror efficiency, detector sensitivity, and any losses due to filters or bad pixels. This is important because not all incoming photons are detected, and the efficiency changes with energy. Therefore, the ARF allows us to correctly estimate the true flux and shape of the source spectrum. In XMM-Newton analysis, the ARF is generated using the `arfgen` task after defining the source and background regions, and it is used together with the RMF during spectral fitting.
Command to generate arf file is given below:

```
arfgen arfset=pn_source.arf spectrumset=pn_source_spec.fits \
withrmfset=yes rmfset=pn_source.rmf \
badpixlocation=pn_clean.fits extendedsource=no \
detmaptype=psf
```

Input: `pn_source_spec.fits`, `pn_source.rmf`, `pn_clean.fits` and create the psf and arf file – `pn_source.arf`

Source spectrum and rmf file used to create the arf file.
After this a very important step come which is backscale.

Step 16: Calculate Backscale

Command to use for backscale:

```
# For source spectrum
backscale spectrumset=pn_source_spec.fits badpixlocation=pn_clean.fits

# For background spectrum
backscale spectrumset=pn_bkg_spec.fits badpixlocation=pn_clean.fits
```

We use the backscale for proper normalization. The relative extraction areas of the source and background regions were calculated using the `backscale` task. This step computes the effective detector area associated with each spectrum while accounting for bad pixels, chip gaps, and region geometry, based on the cleaned event file. The calculated scaling factor is written to the `BACKSCAL` keyword in each spectrum file and is essential for correct background subtraction. Without this scaling, differences in the sizes of the source and background regions would lead to systematic over- or under-subtraction of background counts. By applying `backscale` to both the source (`pn_source_spec.fits`) and background (`pn_bkg_spec.fits`) spectra, the spectra were properly normalized for accurate spectral fitting.

Now, the very imprtant parts come which is to group the data.

Step 17: Group EPIC-pn Spectrum

Grouping the spectrum does not reduce the data itself; rather, it combines neighbouring energy channels to improve the signal-to-noise ratio and make the spectrum suitable for statistical analysis. This step is important because raw spectra often contain bins with very low counts, which can lead to unreliable fitting results.

There are several commonly used tools for grouping:

- **specgroup:** This is the standard SAS task recommended for XMM-Newton data. It performs grouping based on minimum counts and can also control oversampling relative to the instrument resolution. It is generally preferred because it is optimized for XMM-Newton instruments and produces statistically robust spectra.
- **grppha:** This tool, part of HEASoft, was used in this work for grouping the spectra. It provides flexibility in defining custom grouping schemes, such as specifying different bin sizes across energy ranges. This is particularly useful when dealing with varying count rates, where finer binning can be applied at low energies and coarser binning at higher energies. Hence, `grppha` was chosen to allow better control over the binning strategy tailored to the data quality.
- **ftgrouppha (ftgroup):** This is another FTOOLS-based task that allows automated grouping based on criteria such as minimum counts or signal-to-noise ratio. It is useful for scripting and batch processing, but is less commonly used for XMM-Newton data compared to `specgroup`.

In this analysis, `grppha` was adopted due to its flexibility in applying a custom binning scheme suited to the data, while still ensuring sufficient counts per bin for reliable spectral fitting. Final step before XSPEC:

```
specgroup spectrumset=pn_source_spec.fits mincounts=25 \
oversample=3 rmfset=pn_source.rmf arfset=pn_source.arf \
backgndset=pn_bkg_spec.fits groupedset=pn_source_grp.fits
```

```
grppha
pn_source_spec.fits
pn_grp25.pha
group min 25
chkey BACKFILE pn_bkg_spec.fits
chkey RESPFILE pn_source.rmf
chkey ANCRFILE pn_source.arf
exit
```

Note: For my thesis, I have adopted to use the grppha for grouping the data with minimum 25 counts per bin.

2.3.3 RGS data reduction

RGS is a detector which has two parts RGS1 and RGS2. Reduction of both RGS 1 and RGS 2 come with the same command. The data reduction for RGS is same as EPIC-pn till before generating the clean event file. Here we do use the "rgsproc" for get the clean event file for RGS1 and RGS2.

step 1: Generate the clean event file

Here we do use the command:

```
rgsproc
```

In this we would get the clean file for both the detector RGS1 and RGS2. **Here I need to write why we dont need to use the ds9 for source and background region extraction.**

step 2: Combine the RGS 1 and RGS 2

We do combine the data here using the command:

```
rgscombine \
  pha='P0722860201R1S004SRSPEC1001.FIT P0722860201R2S005SRSPEC1001.FIT' \
  bkg='P0722860201R1S004BGSPEC1001.FIT P0722860201R2S005BGSPEC1001.FIT' \
  rmf='P0722860201R1S004RSPMAT1001.FIT P0722860201R2S005RSPMAT1001.FIT' \
  filepha='rgs_combined_source.fits' \
  filebkg='rgs_combined_bkg.fits' \
  filermf='rgs_combined_rmf.fits'
```

As it can be seen that we are putting the pha file, background files and rmf files for both the detector.

step 3: Grouping the data

Like we have mentioned that the grouping can be done in different way like using specgroup, grppha, etc. **For grouping, the same command can be used here, you can go to see the command.**

2.3.4 EPIC-MOS data reduction

: Same here as that till before the generation of clean event files, we follow the same process.

step 1: Generation of clean event file

The command we do use to generate the clean event file for epic-mos data,

```
emchain
```

Write about this here.

step 2:Create light curve for GTI cleaning

with command:

```
evselect table=3411_0830490101_EMOS1_S001_ImagingEvts.ds \
  withrateset=yes rateset=mos1_ltcrv.fits \
  maketimecolumn=yes timebinsize=100 makeratecolumn=yes \
  expression='#XMMEA_EM && (PI>10000 && PI<12000) && (PATTERN==0)'

    evselect table=3411_0830490101_EMOS2_S002_ImagingEvts.ds \
  withrateset=yes rateset=mos2_ltcrv.fits \
  maketimecolumn=yes timebinsize=100 makeratecolumn=yes \
  expression='#XMMEA_EM && (PI>10000 && PI<12000) && (PATTERN==0)'
```

step 3: Checking the light curve

```
  dsplot table=mos1_ltcrv.fits x=TIME y=RATE
dsplot table=mos2_ltcrv.fits x=TIME y=RATE
```

Here, we would identify flare threshold (typically 0.35–0.4 cts/s for MOS). This would be helful to generate the GTI file.

step 4: GTI file creation

create GTI files. Command:

```
tabgtigen table=mos1_ltcrv.fits expression='RATE<=0.10' gtiset=mos1_gti.fits


tabgtigen table=mos2_ltcrv.fits expression='RATE<=0.12' gtiset=mos2_gti.fits
```

step 5: Apply GTI

```
evselect table=3411_0830490101_EMOS1_S001_ImagingEvts.ds \
  withfilteredset=yes filteredset=mos1_clean.fits \
  destruct=yes keepfilteroutput=yes \
  expression='#XMMEA_EM && gti(mos1_gti.fits,TIME) && (PATTERN<=12)'

  evselect table=3411_0830490101_EMOS2_S002_ImagingEvts.ds \
  withfilteredset=yes filteredset=mos2_clean.fits \
  destruct=yes keepfilteroutput=yes \
  expression='#XMMEA_EM && gti(mos2_gti.fits,TIME) && (PATTERN<=12)'
```

step 6: Create Final image

```
evselect table=mos1_clean.fits imagebinning=imageSize \
  imageset=mos1_image.fits withimageset=yes \
  xcolumn=X ycolumn=Y ximagebinsize=80 yimagebinsize=80

  evselect table=mos2_clean.fits imagebinning=imageSize \
  imageset=mos2_image.fits withimageset=yes \
  xcolumn=X ycolumn=Y ximagebinsize=80 yimagebinsize=80
```

step 7: Source and Background extraction using DS9

For the EPIC-MOS detectors, the region of the central chip used in Small Window mode is too small to take a similar approach, so the background was estimated from large regions of blank sky on adjacent chips.

```
evselect table=mos2_clean.fits \
  withspectrumset=yes spectrumset=mos2_src_spec.fits \
  energycolumn=PI spectralbinsize=5 \
  withspecranges=yes specchannelmin=0 specchannelmax=11999 \
  expression='#XMMEA_EM && (PATTERN<=12) && ((X,Y) IN circle(24737,23795,600))'

  evselect table=mos2_clean.fits \
  withspectrumset=yes spectrumset=mos2_bkg_spec.fits \
  energycolumn=PI spectralbinsize=5 \
```

```
  withspecranges=yes specchannelmin=0 specchannelmax=11999 \
  expression='#XMMEA_EM && (PATTERN<=12) && ((X,Y) IN circle(20000,20000,1500))'

  evselect table=mos1_clean.fits \
  withspectrumset=yes spectrumset=mos1_src_spec.fits \
  energycolumn=PI spectralbinsize=5 \
  withspecranges=yes specchannelmin=0 specchannelmax=11999 \
  expression='#XMMEA_EM && (PATTERN<=12) && ((X,Y) IN circle(24737,23795,700))'

  evselect table=mos1_clean.fits \
  withspectrumset=yes spectrumset=mos1_bkg_spec.fits \
  energycolumn=PI spectralbinsize=5 \
  withspecranges=yes specchannelmin=0 specchannelmax=11999 \
  expression='#XMMEA_EM && (PATTERN<=12) && ((X,Y) IN circle(20000,20000,1500))'
```

step 8: Backscale

```
backscale spectrumset=mos1_src_spec.fits badpixlocation=mos1_clean.fits
backscale spectrumset=mos1_bkg_spec.fits badpixlocation=mos1_clean.fits

backscale spectrumset=mos2_src_spec.fits badpixlocation=mos2_clean.fits
backscale spectrumset=mos2_bkg_spec.fits badpixlocation=mos2_clean.fits
```

step 9: Create RMF file

```

rmfgen spectrumset=mos2_src_spec.fits rmfset=mos2.rmf

rmfgen spectrumset=mos1_src_spec.fits rmfset=mos1.rmf
```

step 10: Create ARF File

```

arfgen spectrumset=mos1_src_spec.fits arfset=mos1.arf \
  withrmfset=yes rmfset=mos1.rmf \
  badpixlocation=mos1_clean.fits \
  extendedsource=no detmaptype=psf


  arfgen spectrumset=mos2_src_spec.fits arfset=mos2.arf \
  withrmfset=yes rmfset=mos2.rmf \
  badpixlocation=mos2_clean.fits \
  extendedsource=no detmaptype=psf
```

step 11: Combine MOS1 and MOS2 data

We performed it using the "epicspeccombine",

```
epicspeccombine \
pha="mos1_src_spec.fits mos2_src_spec.fits" \
bkg="mos1_bkg_spec.fits mos2_bkg_spec.fits" \
rmf="mos1.rmf mos2.rmf" \
arf="mos1.arf mos2.arf" \
filepha=mos_combined_src.fits \
filebkg=mos_combined_bkg.fits \
filersp=mos_combined.rsp
```

step 12: Grouping the data

Like we have mentioned that the grouping can be done in different way like using specgroup, grppha, etc. **For grouping, the same command can be used here, you can go to see the command.**

Chapter 3

Spectral Analysis Methodology

Spectral Analysis is the way we do see the pattern that present in source itself. We do get the spectrum in observed frame, so, the spectrum we get has all the effect combined when it reaches to us. To perform the spectral analysis we should have to have the cleaned data after the reduction of raw data which we get from the satellite. Since, we have reduced out the raw data and now we have clean data, therefore, now we can directly proceed with the spectral analysis. To analyse or model the data, we have to have a tool, here we are using the **XSPEC** tool which is commonly used for spectral analysis of data [Arnaud, 1996]. The spectral fitting is performed by minimizing a fit statistic, such as χ^2 for grouped data or the C-statistic for low-count data [Cash, 1979], to determine the best-fit model parameters.
The observed X-ray spectrum can be expressed as:

$$F_{\rm obs}(E) = R(E) \otimes M(E) \tag{3.1}$$

where $M(E)$ represents the intrinsic source model and $R(E)$ denotes the instrumental response (RMF and ARF), and $\otimes$ indicates convolution [Arnaud, 1996]. The observed spectrum is therefore a combination of the intrinsic emission from the source and various modifying effects such as absorption, scattering, and instrumental response.
Since the spectra are observed in the observer's frame, redshift corrections are applied to interpret the physical properties in the rest frame of the source.

Since, we know there are classifications of sources,so, we have to put them, even-though of same class source can show some different feature in spectrum. Therefore to model the data of source, we have different kinds of model for to capture different science. Each model component contains physical parameters such as photon index, temperature, ionization parameter, and column density, which are varied during fitting to match the observed spectrum. Now, we will discuss about the models people use for different science.

3.1 Spectral Fitting Framework

XSPEC, people uses the xspec for spectrum modeling [Arnaud, 1996]. XSPEC has some inbuilt model in it like powerlaw, nthcomp, relxill [Dauser et al., 2016a, García et al., 2014], apec,etc. In XSPEC, models are typically constructed as a combination of multiplicative and additive components, for example:

$$\text{Model} = \text{tbabs} \times \text{ztbabs} \times (\text{continuum} + \text{reflection}) \tag{3.2}$$

It has phenominological and physical model as well inbuild. But people do make the advanced model to capture more physical condition because the inbuild model are not able the complicated signature of source. Therefore, people sometimes prefer to make the **"XSTAR"** [Kallman and Bautista, 2001] which gives highly physical parameters because one make grids based on particular source physical conditions. Grids are just use the atomic database, ionization, velocity of turbulence (particularly for UFOs), column density as in defined range in grids. The XSTAR grids are generated by solving the photoionization balance and radiative transfer in ionized gas. This allows a physically consistent description of ionized absorbers and emitters, where parameters such as ionization parameter, column density, elemental abundances, and redshift can be varied according to the physical conditions of the source. Even XSTAR grid make the physics simple but it is more time consuming at the same time.

3.2 Continuum and Models

The X-ray continuum represents the underlying broad emission from the source, extending over a wide range of energies without sharp features such as lines or edges. In active galactic nuclei (AGNs), this continuum is primarily produced in the immediate vicinity of the supermassive black hole. Optical and ultraviolet photons emitted from the accretion disk are upscattered to X-ray energies by a population of hot electrons in a region known as the corona. This physical process, called inverse Compton scattering, gives rise to a smooth, power-law shaped spectrum in the X-ray band Haardt and Maraschi [1991, 1993].

In general, several radiative processes can contribute to the emission from astrophysical sources. These include thermal emission from hot gas (bremsstrahlung), synchrotron radiation from relativistic electrons in magnetic fields, and Compton scattering processes Rybicki and Lightman [1986]. Thermal bremsstrahlung produces a continuum spectrum that depends strongly on the temperature of the emitting plasma and typically dominates in hot diffuse environments such as galaxy clusters Sarazin [1988]. Synchrotron emission, on the other hand, is important in jet-dominated systems and produces a broad non-thermal spectrum that can extend from radio to X-ray energies Blandford and Königl [1979].

However, in the case of AGNs, especially in the X-ray band, inverse Compton scattering is the dominant mechanism. This is because the accretion disk provides an abundant supply of low-energy (optical/UV) seed photons, while the corona contains high-energy electrons with

temperatures of the order of tens to hundreds of keV. When these photons interact with the hot electrons, they gain energy through repeated scatterings, resulting in a power-law distribution of photon energies Sunyaev and Titarchuk [1980], Haardt and Maraschi [1991]. The efficiency of this process can be understood through the Compton y-parameter:

$$y = \frac{4kT_e}{m_e c^2} \max(\tau, \tau^2) \tag{3.3}$$

where kT_e is the electron temperature, τ is the optical depth of the corona, m_e is the electron mass, and c is the speed of light Rybicki and Lightman [1986]. For $y \sim 1$, Comptonization becomes efficient, leading to the observed X-ray continuum.

The continuum emission can be described in its simplest form by a power-law:

$$F(E) = KE^{-\Gamma} \tag{3.4}$$

where K is the normalization and Γ is the photon index, which determines the slope of the spectrum. Physically, the value of Γ is related to the temperature and optical depth of the coronal electrons responsible for Comptonization Zdziarski et al. [1998].

At higher energies, the spectrum is expected to show a cutoff due to the finite temperature of the electrons in the corona. This can be expressed as:

$$F(E) \propto E^{-\Gamma} \exp\left(-\frac{E}{E_{\text{cut}}}\right) \tag{3.5}$$

where E_{cut} represents the high-energy cutoff, typically related to the electron temperature in the corona Petrucci et al. [2001].

In practice, different models are used to represent the continuum depending on the level of physical detail required. A simple `powerlaw` model provides a phenomenological description of the spectrum, while more physically motivated models such as `nthComp` describe thermal Comptonization of seed photons in a hot plasma Warwick et al. [1996]. The `nthComp` model includes parameters such as the electron temperature and seed photon temperature, allowing a more realistic representation of the coronal emission.

Before reaching the observer, the intrinsic continuum is modified by absorption along the line of sight. In particular, Galactic absorption affects the soft X-ray part of the spectrum and must be accounted for when modeling the data. This effect can be described as:

$$F_{\text{obs}}(E) = F_{\text{int}}(E), e^{-N_{\text{H}}\sigma(E)} \tag{3.6}$$

where N_{H} is the hydrogen column density and $\sigma(E)$ is the photoelectric cross-section Morrison and McCammon [1983], Wilms et al. [2000].

In addition to absorption, a fraction of the primary continuum interacts with surrounding material, such as the accretion disk, producing a reflected component. This reflection modifies the overall continuum shape and introduces additional spectral features, which are discussed in detail in the following subsection George and Fabian [1991], Ross and Fabian [2005].

Thus, the observed X-ray continuum is a combination of intrinsic emission from the corona, modified by absorption and reflection processes. In the following subsections, we describe each of these components in detail, starting with Galactic absorption.

Continuum is the spectra which has all the range of wavelength which we call as the spectrum.

3.2.1 Galactic Absorption

As discussed in the previous section, the intrinsic X-ray continuum emitted from the corona is modified before reaching the observer. One of the most important effects is absorption by neutral material along the line of sight within our own Galaxy, commonly referred to as Galactic absorption. This process primarily affects the soft X-ray part of the spectrum and must be accounted for to correctly recover the intrinsic emission of the source.

Galactic absorption is dominated by photoelectric absorption, where X-ray photons are absorbed by bound electrons in atoms, leading to the ejection of electrons from inner shells. The efficiency of this process depends strongly on energy, with lower-energy photons being more easily absorbed. This results in a suppression of the spectrum at soft X-ray energies. The effect can be described as:

$$F_{\rm obs}(E) = F_{\rm int}(E), e^{-N_{\rm H}\sigma(E)} \tag{3.7}$$

where $F_{\rm int}(E)$ is the intrinsic spectrum, $N_{\rm H}$ is the hydrogen column density along the line of sight, and $\sigma(E)$ is the photoelectric cross-section Morrison and McCammon [1983], Wilms et al. [2000].

The cross-section $\sigma(E)$ is a strong function of energy and contains absorption edges corresponding to different elements such as oxygen, iron, and neon. As a result, Galactic absorption not only reduces the overall flux at low energies but can also introduce subtle features in the spectrum.

In X-ray spectral analysis, this effect is commonly modeled using the `tbabs` model in XSPEC, which incorporates updated interstellar medium abundances and cross-sections Wilms et al. [2000]. The hydrogen column density $N_{\rm H}$ is usually obtained from Galactic surveys and is often fixed during spectral fitting to represent the known absorption along the line of sight.

Correctly accounting for Galactic absorption is essential, as it ensures that the intrinsic continuum shape is accurately determined. Once this foreground absorption is modeled, additional complexities such as intrinsic absorption within the source itself can be investigated, which are discussed in later sections.

3.2.2 Comptonization Continuum

After accounting for Galactic absorption, the dominant component that shapes the X-ray spectrum is the continuum produced by Comptonization in the corona. In this process, low-energy photons originating from the accretion disk interact with a population of hot electrons and gain energy through repeated scatterings. As a result, the initial optical/UV photons are

shifted into the X-ray regime, producing the observed continuum Sunyaev and Titarchuk [1980], Haardt and Maraschi [1991].

The efficiency of this process depends on the temperature and density of the electrons in the corona. A useful quantity to describe this is the Compton y-parameter:

$$y = \frac{4kT_e}{m_e c^2} \max(\tau, \tau^2) \tag{3.8}$$

where kT_e is the electron temperature and τ is the optical depth of the corona Rybicki and Lightman [1986]. When $y \sim 1$, Comptonization becomes efficient and produces a power-law shaped spectrum, which is commonly observed in AGNs.

In spectral analysis, this process is often modeled using the `nthComp` model in XSPEC, which provides a physical description of thermal Comptonization. Unlike a simple power-law, this model includes important parameters such as the electron temperature (kT_e) and the temperature of the seed photons ($kT_{\rm bb}$), allowing a more realistic representation of the continuum Warwick et al. [1996]. The model also naturally produces a high-energy cutoff, which is related to the electron temperature of the corona Petrucci et al. [2001].

The shape of the Comptonized spectrum is therefore controlled by a combination of the photon index, electron temperature, and seed photon temperature. By fitting these parameters, one can gain insight into the physical conditions of the corona, such as its temperature and optical depth.

Thus, Comptonization provides the primary mechanism responsible for the X-ray continuum in AGNs, forming the baseline over which additional spectral components, such as reflection and intrinsic absorption, are superimposed.

3.2.3 Reflection Component

In addition to the primary continuum produced by Comptonization, a significant fraction of the X-ray emission interacts with the surrounding material, particularly the accretion disk, giving rise to a reflected component. This reflection occurs when high-energy photons from the corona illuminate the relatively cold and dense disk, leading to a combination of scattering, absorption, and re-emission processes George and Fabian [1991], Ross and Fabian [2005].

The reflected spectrum is characterized by several distinct features. One of the most prominent is the fluorescent Fe Kα emission line, typically observed around 6.4 keV for neutral iron. This line is produced when inner-shell electrons are ejected and the resulting vacancies are filled by outer-shell electrons George and Fabian [1991]. In addition to line emission, the reflected spectrum also exhibits a broad excess at higher energies (around 20–30 keV), known as the Compton hump, which arises due to multiple Compton scatterings of photons within the disk material Lightman and White [1988].

The shape of the reflection spectrum depends on several physical parameters, including the ionization state of the disk, elemental abundances, and the geometry of the system. The

ionization state is commonly described by the ionization parameter:

$$\xi = \frac{L}{nr^2} \tag{3.9}$$

where L is the illuminating luminosity, n is the gas density, and r is the distance from the ionizing source Tarter et al. [1969]. Higher values of ξ correspond to more highly ionized material, which significantly alters the reflected spectrum.

In X-ray spectral modeling, reflection is commonly described using models such as `xillver` and `relxill`. The `xillver` model calculates the reflection spectrum from an ionized accretion disk, incorporating detailed atomic physics and radiative transfer Garcia et al. [2013]. The `relxill` model further extends this by including relativistic effects, such as gravitational redshift and Doppler broadening, which become important when the reflection originates close to the black hole Dauser et al. [2016b], García et al. [2014].

Thus, the reflection component provides important information about the accretion disk and the environment near the black hole. When combined with the primary continuum, it allows us to probe the geometry, ionization state, and physical conditions of the system. In the following section, we consider additional absorption effects intrinsic to the source, which further modify the observed spectrum.

3.3 Warm Absorber Models

As discussed in the previous sections, the observed X-ray spectrum is shaped not only by the intrinsic continuum and reflection from the accretion disk, but also by absorption processes along the line of sight. While Galactic absorption accounts for foreground material within our own Galaxy, many active galactic nuclei exhibit additional absorption that originates close to the source itself. This intrinsic absorption is often associated with ionized gas in the vicinity of the central engine and is commonly referred to as a *warm absorber* Reynolds [1997].

Warm absorbers consist of partially ionized gas with temperatures typically in the range of 10^4–10^6 K, and are believed to be located at distances ranging from the accretion disk to the narrow-line region. This gas interacts with the outgoing radiation, imprinting absorption features in the X-ray spectrum, particularly in the soft X-ray band. These features include absorption edges and lines from elements such as oxygen, neon, and iron, which provide valuable diagnostics of the physical conditions in the absorbing material Kallman and Bautista [2001].

The physical state of the warm absorber is commonly described using the ionization parameter:

$$\xi = \frac{L}{nr^2} \tag{3.10}$$

where L is the ionizing luminosity, n is the gas density, and r is the distance from the ionizing source Tarter et al. [1969]. This parameter determines the ionization balance of the gas and strongly influences the observed spectral features.

In addition to ionization, the observed absorption features may also be affected by the

velocity of the gas, which can produce shifts in the energy of absorption lines. Such outflows are commonly observed in AGNs and are thought to play an important role in feedback processes between the central black hole and its host galaxy Tombesi et al. [2010b].

Modeling warm absorbers is therefore essential for accurately interpreting the X-ray spectrum and disentangling the effects of intrinsic absorption from the underlying continuum and reflection components. In the following subsections, we describe both phenomenological and physically motivated models used to represent warm absorbers in X-ray spectral analysis.

3.3.1 Semi-physical Model: `zxipcf`

To model the effects of ionized absorption discussed in the previous section, a commonly used approach is the `zxipcf` model in XSPEC. This model provides a practical way to describe partially ionized absorbing material along the line of sight, while still retaining key physical parameters.

The `zxipcf` model is a semi-physical, XSTAR-based absorption model that uses precomputed photoionization grids to represent the interaction of X-ray radiation with ionized gas Reeves et al. [2008]. It includes the main physical quantities that characterize the absorber, such as the column density ($N_{\rm H}$), ionization parameter (ξ), and covering fraction.

The effect of such an absorber on the intrinsic spectrum can be written as:

$$F_{\rm obs}(E) = f, F_{\rm int}(E), e^{-N_{\rm H}\sigma(E,\xi)} + (1-f), F_{\rm int}(E) \tag{3.11}$$

where f is the covering fraction, $N_{\rm H}$ is the column density, and $\sigma(E,\xi)$ represents the energy-dependent cross-section of ionized material.

The covering fraction parameter allows for a partial covering scenario, in which only a fraction of the source emission is absorbed, while the remaining fraction reaches the observer directly. This is particularly useful in AGNs, where the absorbing material may be clumpy or distributed unevenly along the line of sight.

The ionization state of the absorber is described by the ionization parameter:

$$\xi = \frac{L}{nr^2} \tag{3.12}$$

which determines the degree to which the gas is ionized and therefore the strength and position of absorption features Tarter et al. [1969]. Higher values of ξ correspond to more highly ionized gas, which modifies the spectral features accordingly.

Although `zxipcf` is based on physical calculations, it relies on precomputed grids with fixed assumptions about parameters such as turbulence velocity and elemental abundances. Therefore, it is considered a simplified or semi-physical model, providing a balance between physical realism and computational efficiency.

Thus, the `zxipcf` model is widely used to describe ionized absorbers in AGNs, especially when a flexible and computationally efficient representation of the warm absorber is required.

In the next subsection, we discuss fully physical models based on detailed photoionization calculations using XSTAR grids.

3.3.2 Physical Model: XSTAR Photoionization Grids

While the `zxipcf` model provides a convenient and computationally efficient way to describe ionized absorption, a more physically detailed approach can be achieved using photoionization models generated with XSTAR. These models are based on first-principles calculations of the interaction between radiation and ionized gas, allowing a more accurate description of the physical conditions in the absorber.

XSTAR computes the ionization structure of gas exposed to an external radiation field by solving the equations of photoionization equilibrium and radiative transfer Kallman and Bautista [2001]. In this framework, the balance between photoionization and recombination processes determines the population of different ionic species, which in turn produces absorption features in the X-ray spectrum.

The physical state of the gas is characterized by the ionization parameter:

$$\xi = \frac{L}{nr^2} \tag{3.13}$$

where L is the ionizing luminosity, n is the gas density, and r is the distance from the source Tarter et al. [1969]. This parameter controls the ionization balance and strongly influences the resulting spectral features.

In practice, XSTAR is used to generate grids of models by varying key physical parameters such as column density, ionization parameter, elemental abundances, and turbulence velocity. These grids are then imported into XSPEC as table models, allowing direct fitting of observational data.

Compared to simplified models, XSTAR-based grids provide a more realistic representation of ionized absorbers, as they include detailed atomic physics and allow for greater flexibility in describing the physical environment. However, this approach is computationally more intensive and requires careful selection of parameter ranges based on the properties of the source.

Thus, XSTAR photoionization grids offer a powerful and physically robust method for modeling warm absorbers, enabling a deeper understanding of the structure, composition, and dynamics of the ionized gas in AGNs.

Chapter 4

UFO Detection and Statistical Validation

Ultra Fast outflows detections are very difficult because the signature mainly the Fe XXV/Fe XXVI line have the rest transition energy 6.7 and 6.97 keV respectively which lies in the same range where the reflection part lies, therefore detection of UFOs is diffcult. Firslty we need to model the source with different model like nthcomp, importantly reflection model i.e., xillver or relxill varient model, if after the different model, still the feature of absorption line survives then we are nearly sure that the source has the outflows in it

Now once you stablises your continuum model, Now next step comes as the absorption line detection with proper model. TO do this we follow a systematic procedure which we will discuss in details.

4.1 UFO Detection: Blind Line Search

Firslty, is blind search. Blind line search means you use the gaussian model which will detect the lines with significance value. As the UFOs signature appears in the region of 6 to 10 keV. Therefore, we need to check the absoption line using the gaussian model in the range between 6 to 10 keV. So, you add the gaussain (zgauss) model in additive way with continuum because it is additive model and you give the initial paramters line of absorption, width of the gaussain line (sigma), redshift (you freeze it host AGN and now you are doing the line in the rest frame of host galaxy), norm of gaussain (it is negative because absorption is happening). Since, gaussain is searching the line in rest frame of AGN, therefore the line of absorption we get in the model is in the rest frame of AGN and the dip of absorption we get in spectra is in frame of observer. Now with the help of **steppar** command as **steppar <parameter no.> <min. vale> <max. value> <no. of steps>** in **XSPEC** we run the blind line search and it gives us the significance which is χ^2 without line $-$ χ^2 with line. If this difference is more, we would say the detection is there, but how much the detection is, we need to see based on this.

The detection significance can be interpreted using the change in χ^2 ($\Delta\chi^2$), which corresponds to confidence levels as follows:

- $\Delta\chi^2 = 1.0 \rightarrow 68\%$ confidence (1σ)
- $\Delta\chi^2 = 2.71 \rightarrow 90\%$ confidence
- $\Delta\chi^2 = 4.0 \rightarrow 95\%$ confidence (2σ)
- $\Delta\chi^2 = 6.63 \rightarrow 99\%$ confidence
- $\Delta\chi^2 = 9.0 \rightarrow 99.7\%$ confidence (3σ)

Important note: These $\Delta\chi^2$ values are valid for one interesting parameter (for example, line normalization). If more than one parameter is varied, such as both line energy and normalization, then the confidence levels change accordingly, as they depend on the number of degrees of freedom.

Once we got the one line then we would model it with physical model then we would add another gaussian line and again we would search for line in the same range of energy then again we would do the same significance test but now significance would be χ^2 earlier with 1st line $-$ χ^2 of 2nd line.
This process will go on until the absorption line completely captured.

Once we have detected the line. Now the next test would be the statistical test.

4.2 Statistical Methods

In this we would see the robust or significant the absorption line is.

4.2.1 Detection Significance of the Fe K Absorption Line

we employed a hierarchical approach: beginning with the classical F-test, then validating with Monte Carlo simulations, and finally characterizing the full parameter posterior distributions through Markov Chain Monte Carlo (MCMC) analysis

4.2.2 F-test

The most straightforward measure of improvement in spectral fit when adding a new component is the F-test. When a `zgauss` absorption component is added to the baseline continuum model, the fit statistic improves by $\Delta\chi^2$. The F-statistic quantifies whether this improvement is statistically significant relative to the number of additional free parameters introduced:

$$F = \frac{(\chi^2_{\text{base}} - \chi^2_{\text{line}})/\Delta\nu}{\chi^2_{\text{line}}/\nu_{\text{line}}} \tag{4.1}$$

where $\chi^2_{\rm base}$ and $\chi^2_{\rm line}$ are the fit statistics of the baseline and line models respectively, $\Delta\nu = 1$ is the difference in degrees of freedom, and $\nu_{\rm line}$ is the degrees of freedom of the line model. In XSPEC, this is computed as:

```
ftest chi2_line dof_line chi2_base dof_base
```

The F-test returns a null hypothesis probability p, which represents the probability of obtaining an improvement in χ^2 at least as large as observed purely by chance, assuming no real absorption feature is present. A small p-value therefore indicates a statistically significant detection. The corresponding Gaussian significance in units of σ is obtained from:

$$\sigma = \sqrt{2}\,\mathrm{erfinv}(1 - p) \tag{4.2}$$

However, while the F-test provides a useful first estimate, it is known to be unreliable for additive spectral components such as narrow absorption lines, as it assumes the test statistic follows a χ^2 distribution which may break down in the low signal-to-noise regime Protassov et al. [2002]. This motivates the use of Monte Carlo simulations to validate the F-test result.

4.2.3 Monte Carlo Significance

To obtain a more robust estimate of the detection significance that does not rely on asymptotic assumptions, we employed Monte Carlo simulations using the `simftest` command in XSPEC:

```
simftest N_component N_simulations
```

This procedure generates N synthetic spectra drawn from the best-fit baseline model (without the absorption line), fits each synthetic spectrum with both the baseline and line models, and builds an empirical distribution of $\Delta\chi^2$ values. The Monte Carlo p-value is then the fraction of simulations in which the $\Delta\chi^2$ exceeds the observed value — providing a direct, simulation-based estimate of the false positive rate that naturally accounts for the trial factor and non-Gaussian statistics. With the significance of the Fe K feature now established through both classical and simulation-based methods, we proceed to characterize the full posterior distributions of all model parameters through MCMC analysis.

4.2.4 MCMC Analysis

While the F-test and Monte Carlo simulations establish the significance of the detection, they do not fully characterize the uncertainties on the derived parameters or reveal degeneracies between them. To address this, we performed a Markov Chain Monte Carlo (MCMC) analysis using the Goodman–Weare affine-invariant ensemble sampler Goodman and Weare [2010] implemented in XSPEC. The chain was initialized from a Gaussian approximation to the posterior derived from the covariance matrix of the best-fit parameters.
To obtain robust parameter constraints and posterior distributions for all free parameters, we

performed a Markov Chain Monte Carlo (MCMC) analysis using the Goodman-Weare ensemble sampler implemented in XSPEC. The chain was run with 20 walkers, a total length of 10000 steps per walker, and a burn-in period of 5000 steps to ensure convergence. The proposal distribution was initialized using the Gaussian approximation derived from the fit covariance matrix. The final chain contains 10000 samples across free parameters: the warm absorber column density ($N_{\rm H}$), ionization parameter ($\log \xi$), covering fraction ($C_{\rm f}$), and redshift ($z_{\rm abs}$) from the zxipcf/XSTAR grid component; the photon index (Γ) from nthComp; the ionization parameter ($\log \xi$) and normalization from xillverCp; and the line energy (Eline) and normalization from the zgauss component. We used 20 walkers with a chain length of 10,000 steps per walker. The first 5,000 steps were discarded as burn-in to ensure convergence to the stationary distribution. Parameter uncertainties are quoted as the 16th and 84th percentiles of the marginalized posterior distributions, equivalent to 1σ confidence intervals. The resulting posterior distributions were visualized as corner plots Foreman-Mackey [2016], which simultaneously show the marginalized one-dimensional distributions for each parameter and the two-dimensional joint distributions between parameter pairs — revealing any degeneracies that the standard error analysis would miss.

Note: The convergence of three independent statistical methods — the F-test, Monte Carlo simulations, and MCMC posterior analysis — provides a comprehensive framework for assessing the robustness of any spectral detection. The p-value returned by the F-test represents the probability that the observed improvement in fit statistic arises purely from statistical noise under the null hypothesis of no additional spectral component. A smaller p-value therefore indicates a more statistically significant detection. However, since the F-test relies on asymptotic assumptions that may not hold for weak additive components Protassov et al. [2002], Monte Carlo simulations provide a more reliable estimate of the false positive rate by directly sampling the distribution of $\Delta\chi^2$ under the null hypothesis. Finally, while both the F-test and Monte Carlo simulations establish whether a feature is significant, they do not fully characterize the uncertainties on the derived parameters or reveal degeneracies between them. MCMC posterior analysis addresses this by mapping the full joint posterior distribution of all free parameters, providing robust confidence intervals that account for non-Gaussian uncertainties and parameter correlations. Together, these three methods form a self-consistent statistical framework: the F-test provides a rapid initial assessment, Monte Carlo simulations validate the significance without asymptotic assumptions, and MCMC characterizes the parameter space in full — each method building naturally on the limitations of the previous one.

4.3 Velocity Calculation

The outflow velocity of an absorbing medium detected at a redshifted or blueshifted energy in the X-ray spectrum is derived from the relativistic Doppler formula. For an absorption line observed at energy $E_{\rm obs}$ in the detector frame, the rest-frame energy of the transition is $E_{\rm rest}$, and the source is located at cosmological redshift $z_{\rm s}$. The absorber redshift $z_{\rm abs}$ is first determined from

the observed line energy via:

$$1 + z_{\rm abs} = \frac{E_{\rm rest}}{E_{\rm obs}} \tag{4.3}$$

where $E_{\rm rest}$ is the rest-frame energy of the relevant ionic transition — most commonly the Fe XXVI Lyα line at 6.97 keV Tombesi et al. [2010b], Gofford et al. [2013] or the Fe XXV Heα resonance line at 6.70 keV Nardini et al. [2015] for ultrafast outflow detections in the Fe K band. The intrinsic outflow velocity of the absorber in the rest frame of the source is then obtained by correcting for the cosmological redshift of the source Tombesi et al. [2010b]:

$$1 + z_{\rm UFO} = \frac{1 + z_{\rm abs}}{1 + z_{\rm s}} \tag{4.4}$$

where $z_{\rm UFO}$ is the redshift of the absorber in the source rest frame. A blueshifted absorption line corresponds to $z_{\rm UFO} < 0$, indicating outflowing gas moving towards the observer. The outflow velocity is then derived from the relativistic Doppler formula Tombesi et al. [2010b], Reeves et al. [2003]:

$$\frac{v}{c} = \frac{(1 + z_{\rm UFO})^2 - 1}{(1 + z_{\rm UFO})^2 + 1} \tag{4.5}$$

This expression reduces to the classical Doppler formula $v/c \approx -z_{\rm UFO}$ in the non-relativistic limit $|z_{\rm UFO}| \ll 1$, but is required in its full relativistic form for ultrafast outflows where $v/c \gtrsim 0.1$ Tombesi et al. [2013], Pounds et al. [2003]. In the case of a warm absorber detected through its redshift parameter z in an XSTAR photoionization model Kallman and Bautista [2001], the same correction applies — the model redshift z is equivalent to $z_{\rm abs}$ in Equation 4.4, and the outflow velocity is computed via Equation 4.5.

For soft X-ray warm absorbers, which are typically detected at lower ionization and lower outflow velocities than Fe K ultrafast outflows Laha et al. [2021], the same relativistic framework applies. The multi-phase nature of AGN outflows — with a lower-ionization warm absorber in the soft X-ray band and a highly ionized ultrafast component in the Fe K band — has been established in several well-studied sources including PDS 456 Reeves et al. [2020], Nardini et al. [2015], PG 1211+143 Pounds et al. [2003], Reeves et al. [2018], where the soft X-ray and Fe K outflow velocities are found to be consistent, suggesting a common physical origin in a stratified, multi-phase wind launched from the inner accretion disc King and Pounds [2015], Zubovas and King [2012].

In the context of XSTAR photoionization modelling Kallman and Bautista [2001], the ionization parameter ξ is defined as:

$$\xi = \frac{L_{\rm ion}}{n_{\rm H} r^2} \tag{4.6}$$

where $L_{\rm ion}$ is the ionizing luminosity in the 13.6 eV–13.6 keV band, $n_{\rm H}$ is the hydrogen number density of the absorbing gas, and r is the distance from the ionizing source Tarter et al. [1969],

Kallman and Bautista [2001]. The ionization parameter therefore encodes the physical conditions of the absorbing gas, with $\log \xi \sim 1$–2 characterizing the lower-ionization warm absorber phase producing O VII, O VIII, Ne IX, and Mg XI transitions Kaastra et al. [2000], Blustin [2004], while $\log \xi \sim 3$–5 characterizes the highly ionized phase producing Fe XXV and Fe XXVI absorption associated with ultrafast outflows Tombesi et al. [2010b], Gofford et al. [2013], Reeves et al. [2003].

Chapter 5

SMSS J114447.77−430859.3: A Multi-phase Ultrafast Outflow

5.1 Introduction of Source

SMSS J114447.77−430859.3 (hereafter J1144) is a quasi-stellar object at redshift $z = 0.83$, located at a Galactic latitude of $b = +18.1°$, between the constellations of Centaurus and Hydra. It was identified by its optical colours from the SkyMapper Southern Survey (SMSS) during a search for symbiotic binary stars , and reported by Onken et al. [2022] as the most luminous quasar known over the last ~ 9 Gyr of cosmic history. Optical and near-infrared spectroscopy revealed broad Mg II, Hβ, Hα, and Paβ emission lines , from which Onken et al. [2022] measured a black hole mass of $\log_{10}(M_{\rm BH}/{\rm M}_\odot) = 9.4 \pm 0.5$ and a bolometric luminosity of $L_{\rm bol} = (4.7 \pm 1.0) \times 10^{47}$ erg s^{-1}, implying an Eddington ratio of $\lambda_{\rm Edd} \approx 1.4$ — consistent with near- or super-Eddington accretion. J1144 is the optically brightest (unbeamed) quasar known at $z > 0.4$, with $g = 14.5$ mag (AB) and $K = 11.9$ mag (Vega), and has a luminosity approximately eight times greater than 3C 273 Onken et al. [2022].

The first dedicated X-ray study of J1144 was presented by Kammoun et al. [2023], who combined observations from eROSITA/SRG, XMM-Newton, NuSTAR, and Neil Gehrels Swift Observatory to characterize the broadband X-ray properties of the source for the first time. The source shows clear variability by factors of ~ 10 and ~ 2.7 over timescales of a year and of a few days, respectively — remarkably extreme variability for a quasar hosting a black hole of this mass. When fit with an absorbed power law plus high-energy cutoff, the X-ray spectra reveal a photon index $\Gamma = 2.2 \pm 0.2$ and a high-energy cutoff $E_{\rm cut} = 23^{+26}_{-5}$ keV Kammoun et al. [2023]. Assuming Comptonization, Kammoun et al. [2023] estimated a coronal electron temperature of $kT_e = 8$–18 keV and an optical depth of $\tau = 2.5$–5.3 for a slab geometry, making J1144 one of the few quasars where the coronal temperature is directly constrained from the high-energy spectral cutoff. The source was not detected in the ROSAT All-Sky Survey (RASS), but was clearly detected and found to be highly variable across five single passes of the eROSITA All-Sky Survey (eRASS1–5) Kammoun et al. [2023]. The extreme luminosity and near-Eddington accretion rate of J1144, combined with its relatively low redshift compared to other quasars of similar luminosity,

make it an exceptional laboratory for studying the physics of supermassive black hole accretion and the associated feedback mechanisms. In particular, luminous, high-Eddington quasars are the most natural hosts of ultrafast outflows (UFOs), which are thought to be radiatively driven from the inner accretion disc King and Pounds [2015], Zubovas and King [2012] and are detected as blueshifted Fe K absorption features in the X-ray spectrum [Tombesi et al., 2010b, Gofford et al., 2013]. The proximity of J1144 relative to other sources of comparable luminosity — hosted in a galaxy approximately 9.6 billion light years from Earth Kammoun et al. [2023] — allows for high-quality X-ray spectroscopy with XMM-Newton EPIC-pn, making it uniquely suited for a detailed study of its wind properties. The present work reports the first detection of ultrafast outflow signatures in J1144 through a systematic analysis of its two XMM-Newton observations. While Kammoun et al. [2023] established the broadband X-ray continuum properties of J1144 for the first time, their study focused primarily on the characterization of the corona and accretion disc, and did not systematically search for ionized outflow signatures in the X-ray spectrum. This represents a significant gap in our understanding of J1144, given the strong theoretical and observational motivation to expect powerful outflows in such an extreme source.

From a theoretical perspective, radiatively driven winds are expected to be most powerful in sources accreting near or above the Eddington limit King and Pounds [2003, 2015], Zubovas and King [2012], where radiation pressure on highly ionized gas in the inner accretion disc can accelerate material to velocities of $v \sim 0.1$–$0.4c$, producing the blueshifted Fe K absorption features that are the observational hallmark of ultrafast outflows Tombesi et al. [2010b], Gofford et al. [2013]. With an Eddington ratio of $\lambda_{\rm Edd} \approx 1.4$, J1144 sits precisely in this regime. Moreover, the extreme X-ray variability reported by Kammoun et al. [2023] — particularly the factor of ~ 10 variability on year-long timescales — could in part be driven by variations in the column density of an intervening warm absorber or UFO, as has been observed in other variable AGN such as NGC 1365 Risaliti et al. [2005], and PDS 456 Matzeu et al. [2017]. This makes J1144 a natural candidate for harbouring a multi-phase, variable outflow system.

From an observational perspective, the closest analogue to J1144 in the literature is PDS 456 ($z = 0.184$), the most luminous nearby quasar, in which a well-studied multi-phase ultrafast outflow has been detected across the soft X-ray and Fe K bands simultaneously Reeves et al. [2003], Nardini et al. [2015], Reeves et al. [2020]. In PDS 456, the soft X-ray absorber ($\log \xi \sim 3.4$, $v \sim 0.258c$) is found to be consistent in velocity with the Fe K ultrafast outflow Reeves et al. [2020], demonstrating that the two phases are co-spatial and physically connected as parts of a single stratified wind. The striking similarity in luminosity, Eddington ratio, and X-ray spectral slope between J1144 and PDS 456 strongly suggests that J1144 may host a similarly structured multi-phase outflow — but at a luminosity ~ 50 times greater and at a cosmological redshift of $z = 0.83$, making it the most extreme system in which such an outflow study can be conducted with current X-ray facilities.

Furthermore, the two XMM-Newton observations of J1144 available in the archive provide an opportunity to test the temporal consistency of any detected outflow signatures — a critical test for distinguishing real astrophysical features from spurious noise fluctuations. A detection of consistent outflow velocity across two independent epochs, separated by a significant time

interval, would provide strong evidence for a persistent, physically motivated wind rather than a transient spectral artifact Tombesi et al. [2010b].

Motivated by all of the above, the present work undertakes the first systematic search for multi-phase ionized outflows in the XMM-Newton X-ray spectra of J1144, employing photoionization modelling with XSTAR grids Kallman and Bautista [2001] to characterize both the soft X-ray warm absorber and the Fe K ultrafast outflow component independently, and combining F-test significance estimates with MCMC posterior analysis to robustly quantify the detection significance and parameter uncertainties. The detection of a multi-phase outflow in the most luminous quasar of the last 9 Gyr would not only establish J1144 as the highest-luminosity source in which such a wind has been characterized, but would also provide new constraints on the role of AGN feedback in the most extreme accretion environments in the Universe.

Table 5.1: Basic Properties of SMSS J114447.77−430859.3

Source Name	RA (J2000)	Dec (J2000)	Redshift	N_H^{gal} (10^{22} cm^{-2})
SMSS J114447.77−430859.3	176.198987	−43.149834	0.83	0.079

Table 5.2: XMM-Newton EPIC-pn Observation Log of SMSS J114447.77−430859.3

Sr. No.	ObsID	Exposure Time after GTI (ks)	Start Date	End Date
1	0922700101	69.4	2023-06-19	2023-06-21
2	0922700401	94.2	2024-01-13	2024-01-14

5.2 Data reduction

For our purpuse, we have used the epic-pn data. The reduction of these data was also carried out following standard procedures, using the XMM-Newton Science Analysis System (SAS v22.1.0).For the EPIC detectors, we cleaned the raw observation files using EPCHAIN and EMCHAIN for the EPIC-pn detector Strüder et al. [2001], respectively. ource spectra and lightcurves were ex- tracted from the cleaned eventfiles with XMMSELECT using a circular region of radius 30" for source. For the EPIC-pn detector, the background was extracted with 50" from a larger region of blank sky on the same detec- tor chip as the source. and 50" background. The background scaling factor was subsequently computed using the SAS task `backscale`, which calculates the ratio of the effective extraction areas of the source and background regions, accounting for bad pixels, chip gaps, and detector boundaries, ensuring accurate background subtraction during spectral fitting Gabriel et al. [2004] To check for photon pile-up contamination, we used the epatplot task in the XMM-Newton Science Analysis System (SAS). The EPIC data were showed some flare, therefore, we removed somme flare above the threshold (≤ 0.4 cts/s) using GTI. As recommended, we only utilized single and double pat- terned events for EPIC-pn (PATTERN ≤ 4). The necessary instrumental response files for each of the detectors were generated using RMFGEN and ARFGEN. Modeling is done with XSPEC (v22.1.0). We have not used the EPIC-MOS1/MOS2 data for our analysis. And not used RGS data because

net source count rates of combined RGS1+RGS2 are 1.37×10^{-2} cts s^{-1} (0922700101) and 1.82×10^{-2} cts s^{-1} (0922700401), with source fractions of only 18.7% and 39.7% respectively, indicating that both spectra are heavily background-dominated.

5.3 Spectral analysis

We now present a spectral analysis of the 2023-24 observations. We use XSPEC v22.1.0 (Arnaud 1996) to model the data, and quote uncertainties on the spectral parameters at the 90for a single parameter of interest. For grouping the data, we have used the grppha with 25 counts per bin. We fit the EPIC data over the 0.3–10.0 keV band. For galactic absorption column density, we used the HI4PI survey. Source details are present in Table 5.1. This source has the 2 observation IDs above the 40 ks exposure time in XMM-Newton, details present in Table 5.2.

For the modelling, We have started we simple powerlaw model which is phenominological model, next we replaced it with nthcomp which is physical model, takes account for inverse compton physics. And we have seen the feature of reflection too, there in continuum, we have added the xillvercp, now combined the base model we do have:

$$\text{tbabs} \times (\text{nthcomp} + \text{xillverCp}) \tag{5.1}$$

where each component represents a distinct physical process:

- **`tbabs`**: This component accounts for the Galactic photoelectric absorption along the line of sight. It models the attenuation of X-ray photons due to neutral interstellar matter and is parameterized by the hydrogen column density N_{H} [Wilms et al., 2000].
- **`nthcomp`**: This represents the primary X-ray continuum produced by thermal Comptonization.
- **`xillverCp`**: This component models the reflection of the primary continuum from an ionized accretion disk.

Photon index and coronal temperature of xillvercp are tied with nthcomp. Since the norm of nthcomp is tending to zero then we have freezed it to the fitted value.
Inclination was fixed to 30 degree Kammoun et al. [2023]. Abundance of Fe was not constrained, there, abundance frozen to solar (i.e., 1). In Kammoun et al. [2023] mentioned Ecut in range between 28-49 keV, therefore we have fixed the kTe to 20 keV.
Norm of nthcomp was frozen to the best fit.
In the model including `nthComp`, the reflection fraction of `xillverCp` was fixed to -1. According to the `relxill` model documentation Dauser et al. [2016b], setting `refl_frac` $= -1$ instructs the model to return only the reflected component, suppressing the direct continuum emission. This is the physically correct choice when a separate continuum component — in our case `nthComp` — is already included in the model. If `refl_frac` were set to a positive value, `xillverCp` would output both the direct continuum and the reflected emission, resulting in a double-counting of the incident continuum alongside `nthComp`, which would be physically incorrect.

5.4 OBSID: 0922700401

Since we have the baseline model for this source, and we are following this baseline model for all obsids. This baseline model yields a fit statistic of $\chi^2/\mathrm{dof} = 945.42/683$, with a null hypothesis probability of 9.46×10^{-11}, indicating that the baseline continuum alone is a poor description of the data, As shown in Figure 5.1 and that additional spectral complexity is required. The best-fit photon index of $\Gamma = 2.071 \pm 0.007$ is consistent with the steep X-ray continuum expected for a high-Eddington accreting source Kammoun et al. [2023], and the disc ionization parameter of $\log \xi = 3.864 \pm 0.040$ erg cm s^{-1} suggests a highly ionized reflecting medium. The significant residuals remaining after this baseline fit, particularly in the soft X-ray band below 2 keV and in the Fe K band above 6 keV, motivate the inclusion of additional absorption components to model the warm absorber and ultrafast outflow signatures, which are described in the following sections.

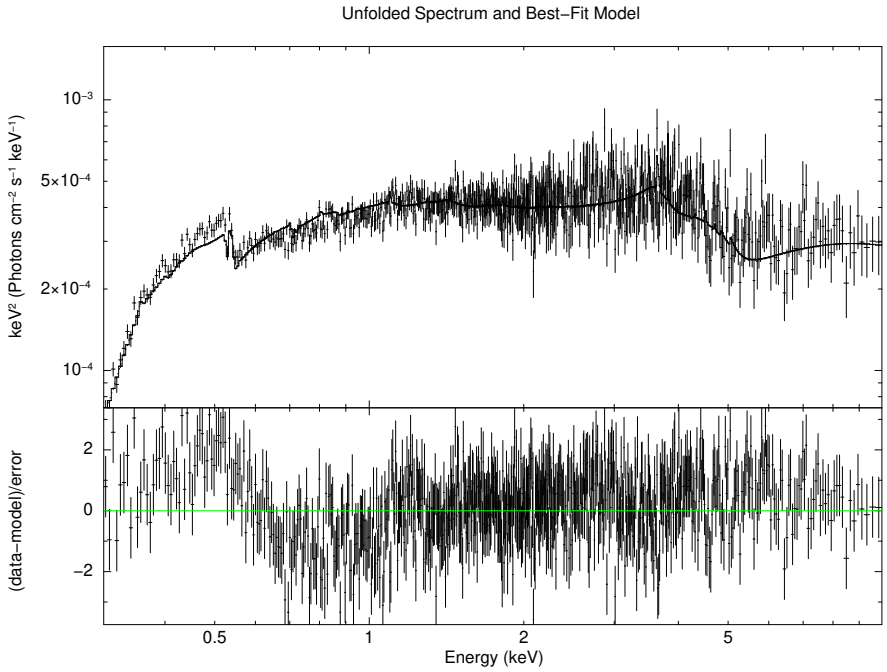


Figure 5.1: Unfolded spectrum and residuals for the baseline model `TBabs*(nthComp+xillverCp)`. Upper panel shows the unfolded spectrum (keV^2 Photons cm^{-2} s^{-1} keV^{-1}) with the best-fit model overlaid. Lower panel shows the residuals in units of (data−model)/error. The poor fit ($\chi^2/\mathrm{dof} = 945.42/683$) is evident from the large systematic residuals, particularly below 2 keV indicating the presence of warm absorption.

5.4.1 Adding the warm absorption

Now, we are adding the "zxipcf" which is semi-physical model for to capture the warm absorption signature.

$$\text{tbabs} \times \text{zxipcf} \times (\text{nthcomp} + \text{xillverCp}) \tag{5.2}$$

`zxipcf` represents a partially covering, ionized absorber modeled using XSTAR photoionization grids, characterized by column density, ionization parameter, and covering fraction. The baseline continuum model leaves significant residuals in the soft X-ray band below 2 keV, indicating the presence of additional absorption along the line of sight. We therefore included a partially covering ionized absorber `zxipcf` [Reeves et al., 2008] to model the warm absorber component. The `zxipcf` model describes a photoionized absorbing medium characterized by a hydrogen column density $N_{\rm H}$, an ionization parameter $\log \xi$, a covering fraction C_f, and a redshift $z_{\rm abs}$. The inclusion of `zxipcf` results in a significant improvement in the fit statistic from $\chi^2/\text{dof} = 945.42/683$ to $\chi^2/\text{dof} = 731.25/679$, corresponding to an improvement of $\Delta\chi^2 = 214.17$ for 4 additional free parameters, as you can see in the Figure 5.2. The null hypothesis probability improves from 9.46×10^{-11} to 8.06×10^{-2}, indicating a statistically acceptable fit. The absorber is characterized by a high column density of $N_{\rm H} = (6.45 \pm 1.40) \times 10^{23}$ cm^{-2} and an ionization parameter of $\log \xi = 3.27 \pm 0.06$ erg cm s^{-1}, with a covering fraction of $C_f = 0.69 \pm 0.10$, consistent with an inhomogeneous wind geometry [Risaliti et al., 2005]. The redshift of the warm absorber was measured at $z_{\rm abs} = 0.449 \pm 0.013$. Despite this improvement, residuals remain in the Fe K band near ~ 5 keV in the observed frame (corresponding to ~ 9 keV in the source rest frame), motivating a further search for a high-velocity Fe K absorption feature, as described in the following section.

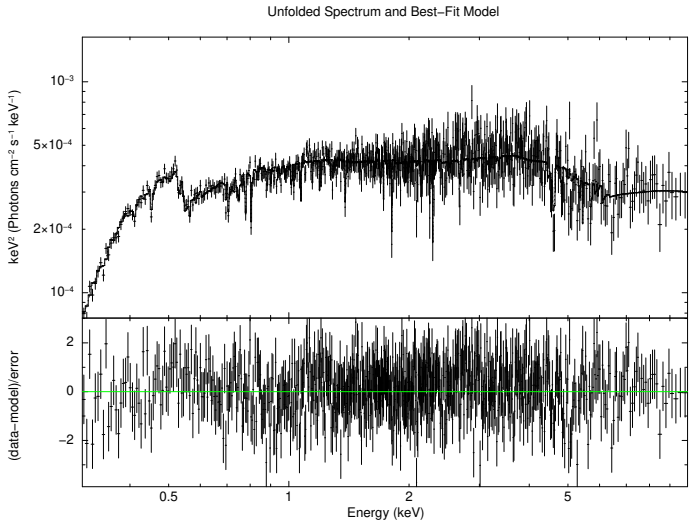


Figure 5.2: Unfolded spectrum and residuals after adding the warm absorber component `zxipcf` to the baseline model `TBabs*zxipcf*(nthComp+xillverCp)`. The fit improves significantly ($\chi^2/\text{dof} = 731.25/679$, $\Delta\chi^2 = 211.27$) compared to the baseline, confirming the presence of ionized warm absorption. A residual dip is visible around 5 keV in the observed frame ($E_{\rm rest} \approx 9.4$ keV), suggesting a possible ultrafast outflow absorption feature.

We have also tried with adding the another zxipcf but the fit was worsen with 2nd zxipcf, therefore we removed it. Now we followed the blind line search for absorption line in range of 6 to 10 keV, so we have added the zgauss.

5.4.2 Adding zgauss for absortion line

Here, after getting the significant results with zxipcf. Now, we need to capture the signature of ultra fast outflows which can be seen in the Figure 5.6.

$$\text{tbabs} \times \text{zxipcf} \times (\text{nthcomp} + \text{xillverCp} + \text{zgauss}) \tag{5.3}$$

`zgauss` is used to model narrow absorption features at a host galaxy redshift, typically with negative normalization, enabling the detection and characterization of absorption lines in the spectrum.

Bline line search: For visual clarity, the spectra were rebinned for plotting purposes using a minimum of 20 counts per bin and a maximum of 5 bins combined. In the Figure 5.3, it is clearly visible in ratio plot that there are consecutive bins in the region from 4.8 to 5.2 keV which is the signature of absorption, not the noise. And apart from this dip, no other dips are significant. In Figure 5.4, when we have added the line using the gaussian, it is clearly visible that the dip around 5.1 keV is clearly captured, means true dip was present at this energy. Then, to see is any other dip is present or not, we have did the systematic search for line using the "steppar" command for line search from 4 to 10 keV, can be seen in Figure 5.5, and it shows on;ly one dip which is at 9.3 keV (this is in the rest frame of AGN).

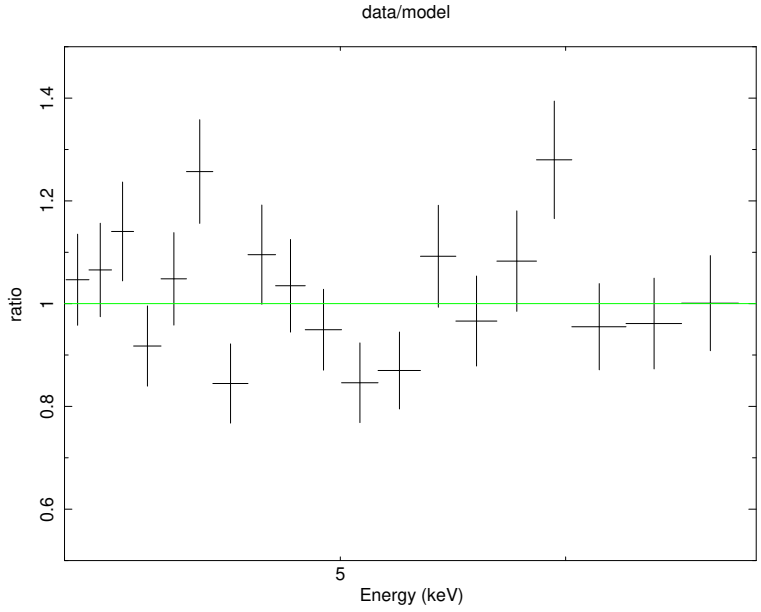


Figure 5.3: Ratio plot with model: tbabs*zixpcf*(nthcomp+xillvercp) and we have not yet added the gaussian line

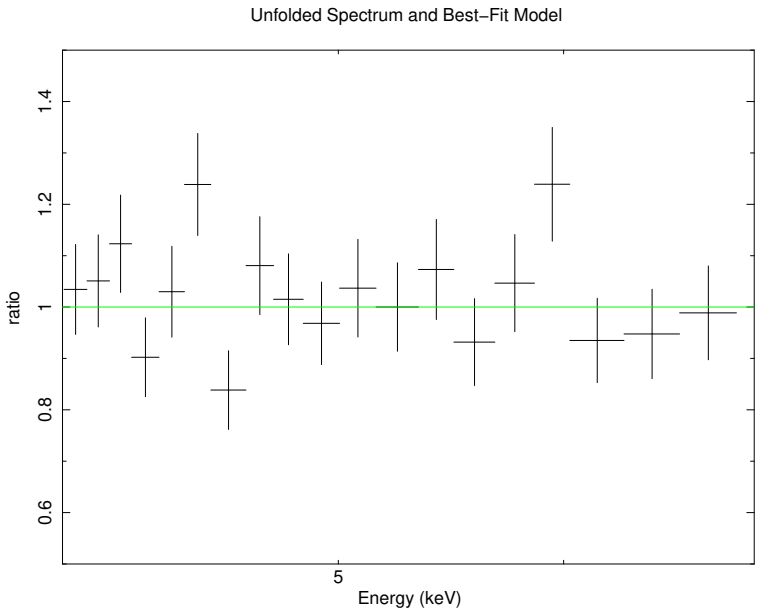


Figure 5.4: Ratio plot with Complete model: tbabs*zxipcf*(nthcomp+xillvercp+zguass), this include the gaussian line for absorption dip, and it is clearly visible that the line is captured around 5.1 keV.

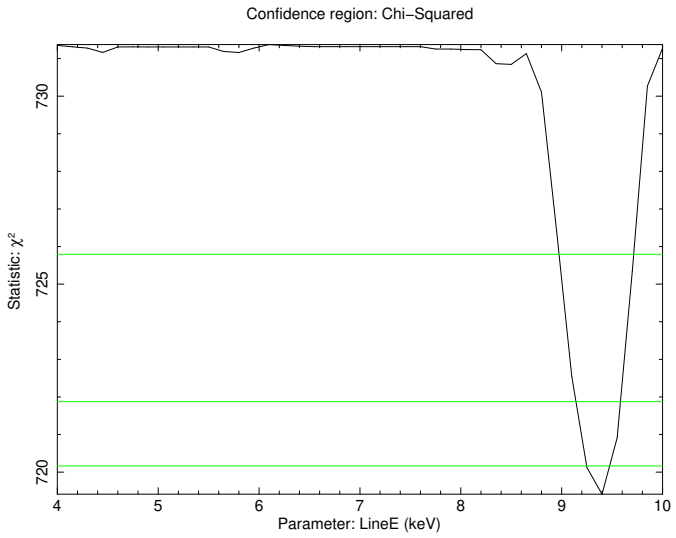


Figure 5.5: We explored how the fit changes when the energy of the absorption line is varied, and the results are shown in Figure X. The three green lines represent different confidence levels: 68%, 90%, and 99%, with the outermost line showing the widest range of possible values. We find a clear best-fit at around 9.3 keV, where all the curves are tightly grouped, indicating that the line energy is well determined. In contrast, there is no clear feature at lower energies (around 4–8 keV), as the fit does not improve in that region. This indicates that there is only one strong absorption feature in the data, and no other statistically significant lines are present.

The ionization parameter of the reflector is poorly constrained, pegging at the hard upper

limit of the `xillverCp` model grid ($\log \xi = 4.62$), with only a lower bound of $\log \xi > 4.37$ obtainable at 90% confidence. This indicates the reflecting material is extremely highly ionized. The residuals remaining near $\sim$ 5.1 keV in the observed frame ($\sim$ 9.4 keV in the source rest frame) after the soft x-ray model motivated the inclusion of an additional Gaussian absorption component `zgauss` to characterize the Fe K absorption feature. The line energy and normalization were left free to vary, with the normalization constrained to be negative to model absorption, while the line width was fixed at $\sigma = 0.1$ keV Tombesi et al. [2010b], Tombesi et al. [2011], consistent with the instrumental resolution of EPIC-pn at these energies [Strüder et al., 2001]. The addition of `zgauss` results in a further improvement in the fit statistic from $\chi^2/\mathrm{dof} = 731.25/679$ to $\chi^2/\mathrm{dof} = 719.22/677$, corresponding to $\Delta\chi^2 = 12.03$ for 2 additional free parameters, as you can see in the Figure 5.6. The null hypothesis probability improves to 1.27×10^{-1}, indicating a good and statistically acceptable description of the data. The best-fit line energy, determined from the 90 per cent confidence interval obtained using the `error` command in XSPEC, is $E_{\mathrm{line}} = 9.392 \pm 0.074$ keV in the source rest frame, corresponding to an observed energy of $\sim$ 5.13 keV in the detector frame. The negative normalization of $-2.16 \times 10^{-6} \pm 0.67 \times 10^{-6}$ photons cm^{-2} s^{-1} confirms the absorption nature of this feature. From zguass, we can't distinguish among Fe XXVI and Fe XXV because zguass only tells line centeriod of energy. Best-fit parameters are present in Table 5.3. outflow velocity corresponding to this line energy is derived in Section 4.3, and the statistical significance of the detection is assessed through F-test and Monte Carlo simulations as described in Section 4.2.

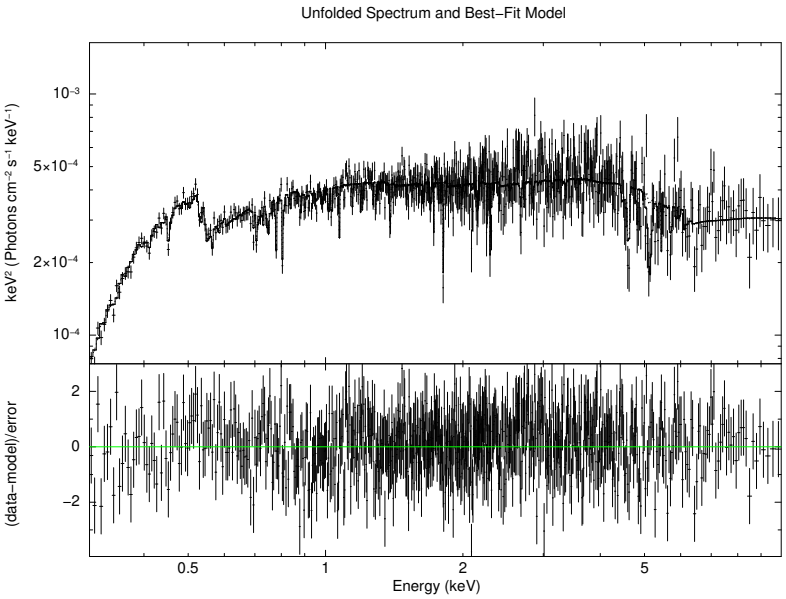


Figure 5.6: Unfolded spectrum and residuals for the final model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)`. The absorption line at $E_{\mathrm{rest}} = 9.4$ keV is now modelled by the `zgauss` component with negative norm, improving the fit to $\chi^2/\mathrm{dof} = 719.22/677$ ($\Delta\chi^2 = 12.03$ compared to the `zxipcf` model). The residuals are now distributed symmetrically around zero across the full 0.3–10 keV band, indicating a statistically acceptable fit with null hypothesis probability of 0.131.

Table 5.3: Best-fit spectral parameters and 90% confidence intervals for the model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)` for ObsID 2 (0922700401) of SMSS J114447.77−430859.3, with the correct Galactic absorption column density $N_H = 7.91 \times 10^{20}$ cm^{-2} [HI4PI Collaboration, 2016]. Errors are quoted at the 90% confidence level ($\Delta\chi^2 = 2.706$).

Component	Parameter	Unit	Value	Status
`TBabs`	N_H	10^{22} cm^{-2}	0.0791	frozen
`zxipcf`	N_H	10^{22} cm^{-2}	$70.119^{+30.939}_{-14.742}$	free
	$\log \xi$	erg cm s^{-1}	$3.257^{+0.082}_{-0.133}$	free
	C_f	–	$0.715^{+0.159}_{-0.130}$	free
	$z_{\rm abs}$	–	$0.449^{+0.019}_{-0.020}$	free
`nthComp`	Γ	–	$2.105^{+0.013}_{-0.019}$	free
	kT_e	keV	20.0	frozen
	$kT_{\rm bb}$	keV	0.05	frozen
	z	–	0.830	frozen
	norm	–	1.34×10^{-22}	pegged[a]
`xillverCp`	Γ	–	$= \Gamma_{\rm nthComp}$	linked
	$A_{\rm Fe}$	solar	1.0	frozen
	kT_e	keV	$= kT_{e,\rm nthComp}$	linked
	$\log \xi$	erg cm s^{-1}	$4.625^{-0.252}_{-0.252}$	pegged[b]
	$\log N$	cm^{-3}	15.0	frozen
	z	–	0.830	frozen
	Incl	deg	30.0	frozen
	refl_frac	–	-1.0	frozen
	norm	10^{-6}	$9.704^{+1.345}_{-0.398}$	free
`zgauss`	$E_{\rm line}$	keV	$9.392^{+0.201}_{-0.266}$	free
	σ	keV	0.1	frozen
	z	–	0.830	frozen
	norm	10^{-6} ph cm^{-2} s^{-1}	$-2.165^{+1.042}_{-1.117}$	free
Fit statistic				
	χ^2/dof	–	719.22/677	–
	χ^2_ν	–	1.062	–
	$p_{\rm null}$	–	1.27×10^{-1}	–
	$\Delta\chi^2$ vs warm absorber	–	12.03 (2 dof)	–

[a] The `nthComp` normalisation is pegged as expected when `refl_frac` $= -1$ in `xillverCp` [Dauser et al., 2016b].

[b] The `xillverCp` $\log \xi$ parameter is pegged at the hard upper limit of the model grid ($\log \xi = 4.70$), with only a lower bound of $\log \xi > 4.37$ obtainable at 90% confidence. This indicates the reflecting material is extremely highly ionized.

The Fe XXVI absorption line energy $E_{\rm line} = 9.392^{+0.201}_{-0.266}$ keV corresponds to an outflow velocity of $v \approx 0.289c$ relative to the systemic redshift $z = 0.831$. The 90% confidence interval on $E_{\rm line}$ spans [9.126, 9.593] keV.

5.4.3 Statistical Tests for Line Detection

F-test and Monte Carlo Significance Test

The statistical significance of the Fe K absorption feature was assessed using two complementary methods. First, the F-test was applied comparing the model with and without the `zgauss` component, yielding an F-statistic of 5.66 with a null hypothesis probability of $p = 3.64 \times 10^{-3}$, corresponding to a significance of $\sim 2.9\sigma$. Second, to account for the known limitations of the F-test for additive line components [Protassov et al., 2002], we performed Monte Carlo simulations using the `simftest` command in XSPEC with 10000 realizations. Out of 10000 simulated spectra drawn from the baseline model without the absorption line, only 113 produced a $\Delta\chi^2$ improvement as large as or larger than observed, yielding a Monte Carlo null hypothesis probability of $p = 1.13 \times 10^{-2}$, corresponding to a significance of $\sim 2.3\sigma$. The consistency between the F-test and Monte Carlo significance estimates confirms the robustness of the detection.

Table 5.4: Statistical significance of the Fe K absorption line detection in SMSS J114447.77−430859.3 (ObsID 0922700401).

Test	Value	Significance
F-test	$F = 5.66$, $p = 3.64 \times 10^{-3}$	$\sim 2.9\sigma$
MC `simftest` (10000 trials)	$p = 1.13 \times 10^{-2}$	$\sim 2.3\sigma$

Equivalent Width

Please find the details in Table 5.5.

Table 5.5: Equivalent width of the Fe K absorption feature in SMSS J114447.77−430859.3 (ObsID 0922700401). Errors are quoted at the 90 per cent confidence level derived from the MCMC chain.

Feature	EW (eV)	90% CI (eV)
Fe XXVI Lyα	$-68.7^{+19.0}_{-23.0}$	-91.7 to -49.7

The negative sign confirms it is an absorption line. The magnitude of 68.7 eV is a reasonably strong absorption feature, consistent with UFO detections in the literature (typically EW $\sim$ 20–100 eV for Fe K UFOs). The equivalent width of the Fe K absorption feature was measured using the `eqwidth` command in XSPEC, with uncertainties derived from the MCMC chain using the loaded chain file. The best-fit equivalent width is EW $= -68.7^{+19.0}_{-23.0}$ eV, where the negative sign confirms the absorption nature of the feature. The 90 per cent confidence interval spans -91.7 to -49.7 eV — entirely negative — confirming that a zero equivalent width is ruled out at the 90 per cent confidence level, and that the feature is a robust absorption detection rather than a statistical fluctuation. The measured equivalent width is consistent with the typical range of Fe XXVI Lyα absorption features detected in ultrafast outflow systems [Tombesi et al., 2010b], where Fe K UFO equivalent widths typically span $\sim$30–200 eV, placing J1144 firmly within this population.

5.4.4 MCMC Analysis

To obtain robust parameter constraints and posterior distributions for all free parameters, we performed a Markov Chain Monte Carlo (MCMC) analysis using the Goodman-Weare ensemble sampler implemented in XSPEC. The chain was run with 20 walkers, a total length of 10000 steps per walker, and a burn-in period of 5000 steps to ensure convergence. The proposal distribution was initialized using the Gaussian approximation derived from the fit covariance matrix. The final chain contains 10000 samples across 9 free parameters: the warm absorber column density (N_H), ionization parameter ($\log \xi$), covering fraction (C_f), and redshift ($z_{\rm abs}$) from the `zxipcf` component; the photon index (Γ) from `nthComp`; the ionization parameter ($\log \xi$) and normalization from `xillverCp`; and the line energy ($E_{\rm line}$) and normalization from the `zgauss` component.

The resulting posterior distributions and two-dimensional confidence contours for all free parameters are shown in Figure 5.7. The diagonal panels show the marginalized one-dimensional posterior histograms, with dashed vertical lines marking the 16th, 50th, and 84th percentiles. The off-diagonal panels show the two-dimensional joint posteriors at the 68%, 90%, and 99% confidence levels. The red lines and points mark the best-fit values from the χ^2 minimization.

The MCMC analysis confirms robust constraints across all model components. The warm absorber redshift is among the most tightly constrained parameters, with $z_{\rm abs} = 0.448^{+0.012}_{-0.015}$, corresponding to an outflow velocity of $v \approx 0.230c$ relative to the systemic redshift of $z = 0.831$. The photon index is similarly well constrained at $\Gamma = 2.100^{+0.008}_{-0.009}$, consistent with a moderately steep power-law continuum typical of luminous AGN. The warm absorber ionization parameter and covering fraction are recovered at $\log \xi_{\rm zxipcf} = 3.231^{+0.066}_{-0.060}$ and $C_f = 0.709^{+0.078}_{-0.075}$, respectively. The reflector ionization is high at $\log \xi_{\rm xillver} = 4.574^{+0.070}_{-0.116}$, with normalization $\mathrm{norm}_{\rm xillver} = 9.997^{+0.765}_{-0.478} \times 10^{-6}$. The absorption line energy is recovered at $E_{\rm line} = 9.375^{+0.167}_{-0.154}$ keV, with the normalization robustly negative at $\mathrm{norm}_{\rm zgauss} = -1.989^{+0.631}_{-0.733} \times 10^{-6}$, confirming the absorption nature of the feature from the MCMC posteriors alone. The column density $N_H = 78.960^{+15.999}_{-13.127} \times 10^{22}$ cm^{-2} shows a somewhat skewed posterior distribution, which is typical for partial covering absorber models where N_H and C_f are partially degenerate, as visible in their joint posterior panel. This skewed feature is also visible in the contour plot (Figure 5.7).

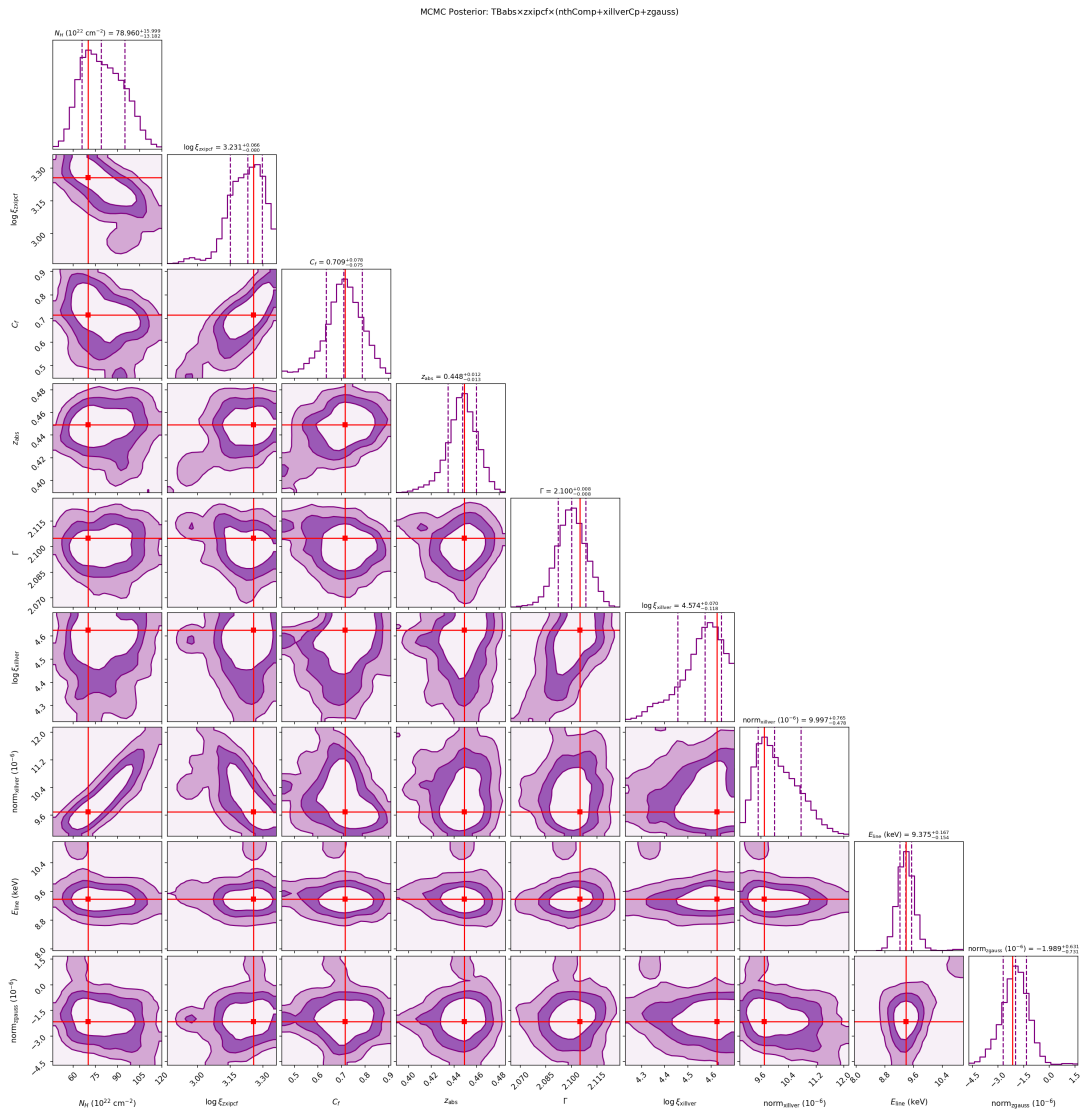


Figure 5.7: MCMC posterior distributions for all free parameters of the model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)`. Diagonal panels show marginalized 1D posteriors, off-diagonal panels show 2D joint posteriors at 68%, 90%, and 99% confidence. Red lines mark best-fit values.

5.4.5 Soft X-ray ultrafast Outflows and Fe K line ultrafast outflows velocity calculations

5.4.6 Soft X-ray ultrafast Outflows

Substituting the redshift obtained from the `zxipcf` model [Kallman and Bautista, 2001, Kallman et al., 2004] $z_{\rm abs} = 0.449^{+0.019}_{-0.020}$ into the relativistic Doppler formula:

$$1 + z_{\rm UFO} = \frac{1 + z_{\rm abs}}{1 + z_{\rm source}} = \frac{1 + 0.449}{1 + 0.830} = \frac{1.449}{1.830} = 0.7918 \tag{5.4}$$

$$\frac{v_{\rm out}}{c} = \frac{(0.7918)^2 - 1}{(0.7918)^2 + 1} = \frac{0.6269 - 1}{0.6269 + 1} = \frac{-0.3731}{1.6269} \approx -0.230c \tag{5.5}$$

Propagating the uncertainty on $z_{\rm abs}$ $(^{+0.019}_{-0.020})$:

- $z_{\rm abs,min} = 0.429 \Rightarrow 1 + z_{\rm UFO} = 1.429/1.830 = 0.7809 \Rightarrow v = 0.242c$
- $z_{\rm abs,max} = 0.468 \Rightarrow 1 + z_{\rm UFO} = 1.468/1.830 = 0.8022 \Rightarrow v = 0.217c$

The soft X-ray ultra-fast outflow velocity is therefore:

$$v_{\rm soft} = 0.230^{+0.012}_{-0.013}c \tag{5.6}$$

The gas is outflowing towards the observer at $v \approx 0.230c$ relative to the source redshift $z_{\rm source} = 0.830$ [Onken et al., 2022], placing it firmly in the ultra-fast outflow regime ($v \geq 0.1c$; Tombesi et al. [2010b], Reeves et al. [2020]).

Fe K line ultrafast outflows

As we have got the $E_{\rm rest}$ from the best fit and now substituting the best-fit line energy $E_{\rm rest} = 9.392^{+0.201}_{-0.266}$ keV (90% CI: [9.126, 9.593] keV) and identifying the feature as Fe XXVI Lyα ($E_{\rm ion} = 6.97$ keV) [Tombesi et al., 2010b, Gofford et al., 2013] into the equation:

$$E_{\rm rest} = E_{\rm obs} \times (1 + z_{\rm source}) \tag{5.7}$$

where $E_{\rm obs}$ is the observed line energy in the spectrum and $z_{\rm source} = 0.830$ is the source redshift. The blueshift parameter $z_{\rm UFO}$ is then defined as:

$$1 + z_{\rm UFO} = \frac{E_{\rm ion}}{E_{\rm rest}} \tag{5.8}$$

where $E_{\rm ion}$ is the rest-frame energy of the identified ion. The outflow velocity is then obtained from the relativistic Doppler formula:

$$\frac{v_{\rm out}}{c} = \frac{(1 + z_{\rm UFO})^2 - 1}{(1 + z_{\rm UFO})^2 + 1} \tag{5.9}$$

$$1 + z_{\rm UFO} = \frac{6.97}{9.392} = 0.7421 \tag{5.10}$$

$$\frac{v_{\rm out}}{c} = \frac{(0.7421)^2 - 1}{(0.7421)^2 + 1} = \frac{0.5507 - 1}{0.5507 + 1} = \frac{-0.4493}{1.5507} \approx -0.290c \tag{5.11}$$

Propagating the 90% confidence interval on $E_{\rm line}$ ([9.126, 9.593] keV):

- $E_{\rm rest,min} = 9.126$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.97/9.126 = 0.7638 \Rightarrow v = 0.263c$
- $E_{\rm rest,max} = 9.593$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.97/9.593 = 0.7261 \Rightarrow v = 0.309c$

The UFO outflow velocity is therefore:

$$v_{\rm UFO} = 0.290^{+0.019}_{-0.027}c \tag{5.12}$$

The negative sign confirms a blueshift, i.e. the absorbing gas is outflowing towards the observer at $v_{\rm UFO} \approx 0.290c$, consistent with an ultrafast outflow (UFO), defined as $v > 0.1c$ [Tombesi et al., 2010b].

Alternatively, if the absorption feature is identified as Fe XXV Heα ($E_{\rm ion} = 6.70$ keV), the implied outflow velocity is higher:

$$1 + z_{\rm UFO} = \frac{6.70}{9.392} = 0.7134 \tag{5.13}$$

$$\frac{v_{\rm out}}{c} = \frac{(0.7134)^2 - 1}{(0.7134)^2 + 1} = \frac{0.5089 - 1}{0.5089 + 1} = \frac{-0.4911}{1.5089} \approx -0.325c \tag{5.14}$$

Propagating the 90% confidence interval on $E_{\rm line}$ ([9.121, 9.599] keV):

- $E_{\rm rest,min} = 9.126$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.70/9.126 = 0.7341 \Rightarrow v = 0.300c$
- $E_{\rm rest,max} = 9.593$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.70/9.593 = 0.6980 \Rightarrow v = 0.344c$

The Fe XXV identification therefore implies:

$$v_{\rm UFO} = 0.325^{+0.019}_{-0.025}c \tag{5.15}$$

The two possible identifications of the absorption feature admit outflow velocities of $v \approx 0.290c$ (Fe XXVI Lyα, $E_{\rm ion} = 6.97$ keV) and $v \approx 0.325c$ (Fe XXV Heα, $E_{\rm ion} = 6.70$ keV). To determine which ion is responsible for the absorption, one must consider the ionization state of the absorbing gas. The dominant iron ion is determined by the ionization parameter $\log \xi$: Fe XXV (He-like iron) dominates at $\log \xi \lesssim 3.5$ erg cm s^{-1}, while Fe XXVI (H-like iron) dominates at $\log \xi \gtrsim 4.0$ erg cm s^{-1} [Kallman and Bautista, 2001, Tombesi et al., 2011]. In the intermediate range $3.5 \lesssim \log \xi \lesssim 4.0$ erg cm s^{-1}, both ions coexist and contribute comparably to the Fe K absorption trough, making an unambiguous identification from the line centroid energy alone impossible [Reeves et al., 2009, Tombesi et al., 2011].

5.5 OBSID: 0922700101

We have started with two binning scheme i.e., 25 counts/bin and 15 counts/bin. And we also the results and found that the fe line is robust at 5 keV. Therefore, in final, we have followed 25 counts/s binning scheme.

This baseline model yields a fit statistic of $\chi^2/{\rm dof} = 717.45/646$, with a null hypothesis probability of 2.64×10^{-2}, indicating that the baseline continuum alone is a poor description of the data and that additional spectral complexity is required. The best-fit photon index of $\Gamma = 1.906 \pm 0.007$ is consistent with the steep X-ray continuum expected for a high-Eddington accreting source [Kammoun et al., 2023], and the disc ionization parameter of $\log \xi = 3.940 \pm 0.053$ erg cm s^{-1} suggests a highly ionized reflecting medium. The significant residuals remaining after this baseline fit, particularly in the soft X-ray band below 2 keV and in the Fe K band above 6 keV can be seen in the (Figure 5.8), motivate the inclusion of additional absorption components to

model the warm absorber and ultrafast outflow signatures, which are described in the following sections.

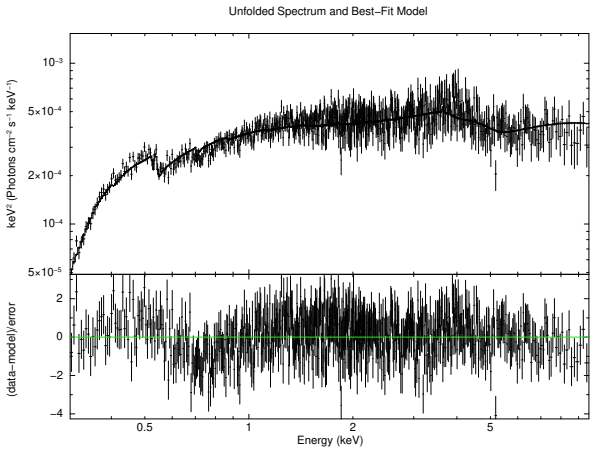


Figure 5.8: Unfolded spectrum and residuals for the baseline model `TBabs*(nthComp+xillverCp)`. Upper panel shows the unfolded spectrum (keV^2 Photons cm^{-2} s^{-1} keV^{-1}) with the best-fit model overlaid. Lower panel shows the residuals in units of (data−model)/error. The poor fit (χ^2/dof = 717.45/646) is evident from the large systematic residuals, particularly below 2 keV indicating the presence of warm absorption.

5.5.1 Adding the warm absorption: Zxipcf

The baseline model provides a statistically acceptable fit to the spectrum, with $\chi^2/\nu = 668.52/643 = 1.039$ and null hypothesis probability $p = 0.235$. The warm absorber column density $N_H = 64.872 \pm 28.196 \times 10^{22}$ cm^{-2}, ionization parameter $\log \xi = 3.423 \pm 0.051$ erg cm s^{-1}, and covering fraction $C_f = 0.666 \pm 0.160$ are all broadly consistent within their (large) uncertainties with the corresponding 092270401 values of $N_H = 70.119^{+30.939}_{-14.742} \times 10^{22}$ cm^{-2}, $\log \xi = 3.257^{+0.082}_{-0.133}$ erg cm s^{-1}, and $C_f = 0.715^{+0.159}_{-0.130}$. The photon index $\Gamma = 1.937 \pm 0.015$ is also consistent with the 092270401 value of $\Gamma = 2.105^{+0.013}_{-0.019}$ within $\sim 2\sigma$, suggesting no dramatic change in the underlying continuum shape between the two epochs. We note that the reflector ionization $\log \xi_{\rm xillver} = 4.403 \pm 0.209$ is somewhat lower than in 092270401 ($4.625^{-0.252}_{-0.252}$), though still consistent within errors given the large uncertainties on both measurements. Please find the (Figure 5.9) for details.

On the choice to freeze the warm absorber redshift. On the choice to freeze the warm absorber redshift, we note that allowing the `zxipcf` redshift to vary freely in ObsID 0922700101 results in a best-fit value of $z_{\rm abs} = 0.371 \pm 0.010$, corresponding to an outflow velocity of $v \approx 0.277c$. This differs from the value obtained in ObsID 092270401 ($z_{\rm abs} = 0.448$, $v \approx 0.230c$). However, the improvement in the fit statistic is modest ($\Delta\chi^2 = 4.25$ for one additional free parameter), corresponding to a significance of only $\sim 2\sigma$, which is not statistically compelling [Press et al., 1992] In addition, the large uncertainties on the covering fraction ($C_{\rm f} = 0.726 \pm 0.214$) and column density ($N_{\rm H} = 64.03 \pm 41.67 \times 10^{22}$ cm^{-2}) indicate that the warm absorber is not well

constrained in this observation. This is consistent with known degeneracies between absorber parameters in CCD-resolution X-ray spectra [Reynolds, 1997, Kallman and Bautista, 2001, Tombesi et al., 2013]. Given these limitations, we adopt a conservative approach by fixing the absorber redshift at $z_{\rm abs} = 0.448$, as obtained from ObsID 092270401. This choice is supported both by the weak statistical preference for a different value and by the physical expectation that the soft X-ray warm absorber traces the same outflowing wind in both epochs [Parker et al., 2017, Matzeu et al., 2017]. This approach allows for a more consistent and reliable comparison of the Fe XXVI absorption properties between the two observations, which is the primary focus of the analysis. A dedicated higher-resolution soft X-ray analysis – for example using the RGS data or future *NewAthena* observations – would be required to robustly constrain the warm absorber velocity independently in each epoch and assess whether a genuine velocity shift has occurred in the soft X-ray wind between June 2023 and January 2024.

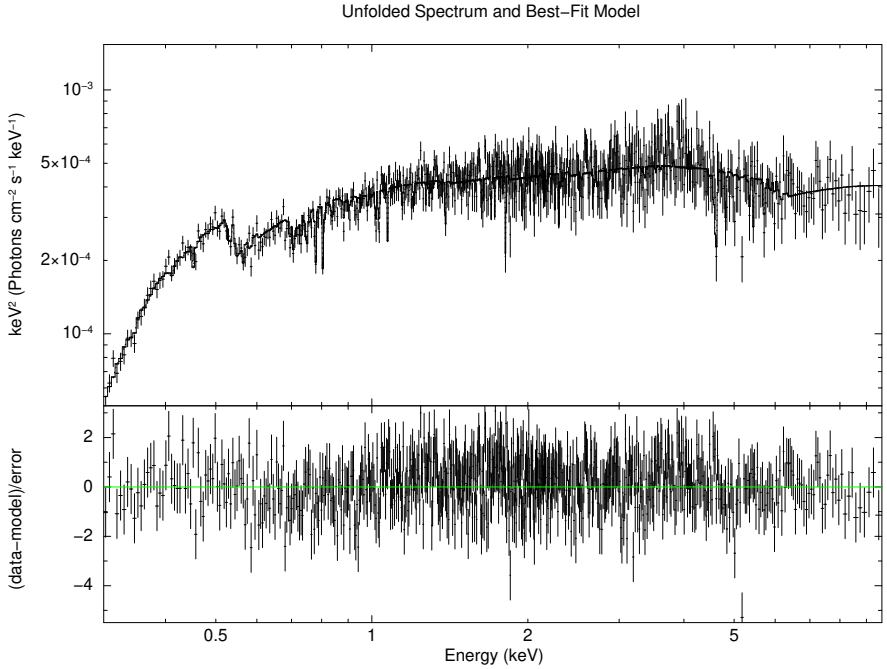


Figure 5.9: Unfolded spectrum and residuals after adding the warm absorber component `zxipcf` to the baseline model `TBabs*zxipcf*(nthComp+xillverCp)`. The fit improves significantly (χ^2/dof = 667.25/644) compared to the baseline, confirming the presence of ionized warm absorption.

5.5.2 Blind Gaussian Line Scan

To search for Fe K absorption features in 0922700101 without assuming a prior line energy, we performed a blind Gaussian scan over the observed-frame energy range 4.0–10.0 keV using the `steppar` command with 40 logarithmically spaced steps. The line width was frozen at $\sigma = 0.1$ keV Tombesi et al. [2010b], Tombesi et al. [2011], consistent with the EPIC-pn instrumental resolution at ~ 9 keV and with the 0922700401 analysis. Prior to the scan, we examined the ratio plot

of the data to the continuum model `TBabs*zxipcf*(nthComp+xillverCp)` in the Fe K band (Figure 5.10). A coherent multi-bin deficit is visible around ~ 5 keV observed frame, corresponding to ~ 9.36 keV rest frame at $z = 0.83$, consistent with Fe XXVI Lyα absorption at $v \approx 0.289c$.

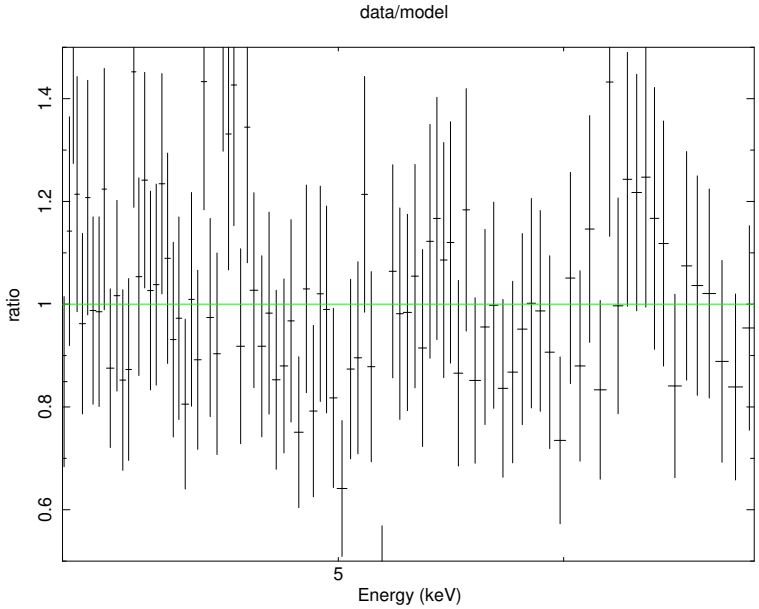


Figure 5.10: This is with model: tbabs*zixpcf*(nthcomp+xillvercp) only, here's not yet gaussian model added.

The blind Gaussian scan reveals a single, well-defined minimum at $E_{\rm line} = 9.40$ keV with $\Delta\chi^2 = 24.96$ relative to the continuum baseline (Figure 5.11). No other energy in the scanned range approaches this significance – the next deepest feature is at ~ 6.1 keV with $\Delta\chi^2 = 2.6$, which is entirely consistent with noise given the number of trials. The minimum is narrow, symmetric, and located at exactly the same energy as the Fe XXVI absorption feature detected in 0922700401 ($E_{\rm line} = 9.392$ keV).

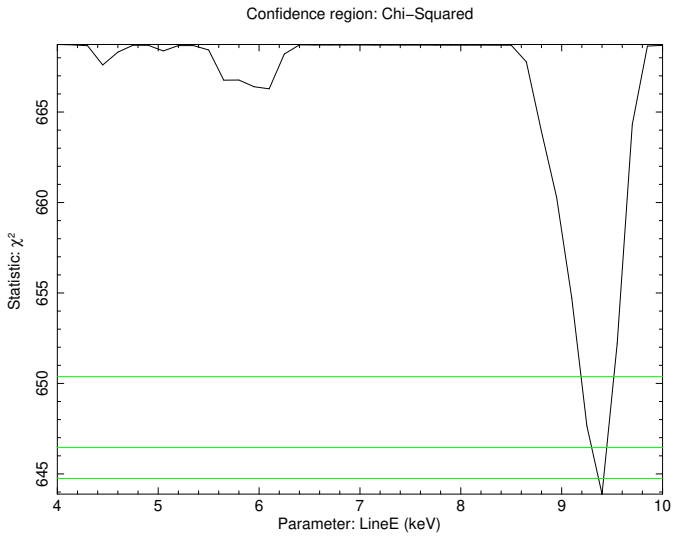


Figure 5.11: Blind line search plot with the gassian model using the steppar from 6 to 10 keV.

Adding the `zgauss` component at the best-fit energy and allowing the normalization to vary with negative hard limits, the fit improves to $\chi^2/\nu = 643.53/642 = 1.002$, with the line normalization converging to norm $= -3.441 \times 10^{-6} \pm 7.440 \times 10^{-7}$ ph cm^{-2} s^{-1}. The Normalized residuals plot with the line included (Figure 5.12) shows the deficit around ~ 5.1 keV is substantially reduced, confirming that the Gaussian component is absorbing a genuine spectral feature rather than fitting noise. The best-fit line energy $E_{\rm line} = 9.364 \pm 0.053$ keV is fully consistent with the ObsID 2 value of 9.392 ± 0.074 keV within the measurement uncertainties.

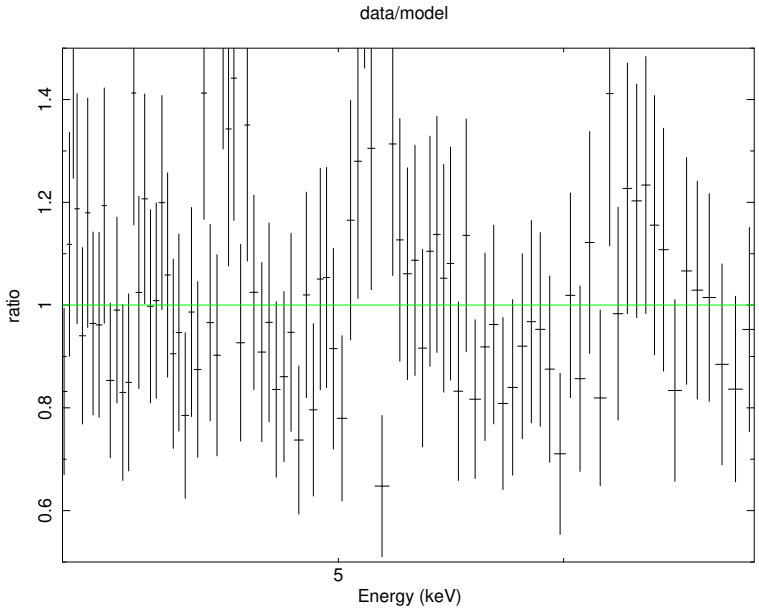


Figure 5.12: Ratio plot with Complete model: tbabs*zxipcf*(nthcomp+xillvercp+zguass), gaussain line included.

Figure 5.10 — Ratio plot: Looking at the 4.5–7 keV observed frame region — there are 2–3 consecutive bins dipping below 0.5 around $\sim$ 5 keV (observed frame = 9.4 keV rest frame). This is a coherent multi-bin dip, not a single bin fluctuation. This is exactly what a real absorption line looks like.

Figure 5.11 — Steppar contour plot: This is the most important. The minimum is at 9.4 keV, deep and clean — going well below the green line (which marks the minimum χ^2). The profile drops sharply only around 9.4 keV and nowhere else in the band. This is a real line signature.

Figure 5.12 — Ratio plot with gaussian line: The dip around 5 keV is significantly reduced when the `zgauss` is included — the residuals become flatter. This means the Gaussian is genuinely absorbing a real spectral feature, not just fitting noise.

Final model plot: **Final model plot:**

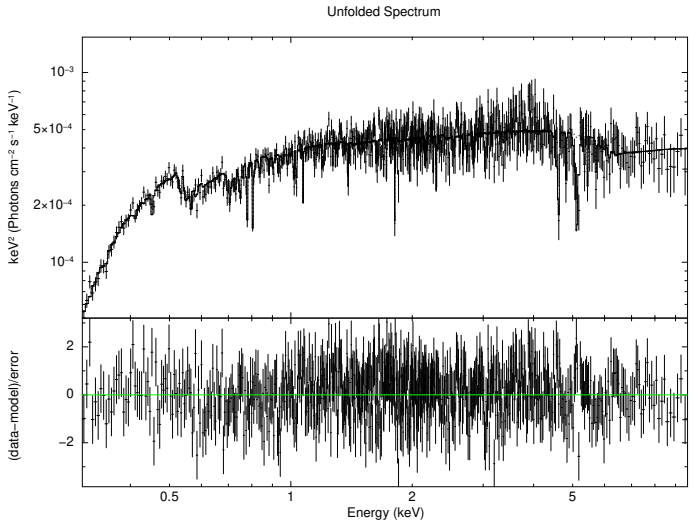


Figure 5.13: Unfolded spectrum and residuals for the final model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)`. The absorption line at $E_{\rm rest} = 9.4$ keV is now modelled by the `zgauss` component with negative norm, improving the fit to $\chi^2/\mathrm{dof} = 719.22/677$ ($\Delta\chi^2 = 12.03$ compared to the `zxipcf` model). The residuals are now distributed symmetrically around zero across the full 0.3–10 keV band, indicating a statistically acceptable fit with null hypothesis probability of 0.131.

Caveat on significance. The raw $\Delta\chi^2 = 24.96$ corresponds to a formal significance for 2 additional free parameters (line energy and normalization). However, this does not account for the look-elsewhere effect arising from scanning 40 independent energy bins across the Fe K band. Correcting for the number of trials reduces the effective significance. Furthermore, we note that the warm absorber parameters in 0922700101 are less well constrained than in 0922700401 due to the shorter exposure time, introducing additional continuum uncertainty that may partially contribute to the apparent $\Delta\chi^2$ improvement. We therefore classify this as a *tentative* detection of the Fe XXVI UFO in 0922700101, with a significance that we conservatively estimate at $\sim$ 3–4σ

accounting for these systematics. A dedicated Monte Carlo significance assessment using `fakeit` simulations would be required to establish the precise corrected significance, which we defer to future work.

Table 5.6: Best-fit spectral parameters and 90% confidence intervals for the model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)` for ObsID 1 (0922700101) of SMSS J114447.77−430859.3, with the correct Galactic absorption column density $N_H = 7.91 \times 10^{20}$ cm^{-2} [HI4PI Collaboration, 2016]. Errors are quoted at the 90% confidence level ($\Delta\chi^2 = 2.706$).

Component	Parameter	Unit	Value	Status
`TBabs`	N_H	10^{22} cm^{-2}	0.0791	frozen
`zxipcf`	N_H	10^{22} cm^{-2}	$69.737^{+34.102}_{-16.632}$	free
	$\log \xi$	erg cm s^{-1}	$3.428^{+0.054}_{-0.106}$	free
	C_f	–	$0.769^{+0.176}_{-0.176}$	free
	$z_{\rm abs}$	–	0.448	frozen
`nthComp`	Γ	–	$1.943^{+0.018}_{-0.024}$	free
	kT_e	keV	20.0	frozen
	$kT_{\rm bb}$	keV	0.05	frozen
	z	–	0.830	frozen
	norm	–	2.865×10^{-21}	pegged[a]
`xillverCp`	Γ	–	$= \Gamma_{\rm nthComp}$	linked
	$A_{\rm Fe}$	solar	1.0	frozen
	kT_e	keV	$= kT_{e,\rm nthComp}$	linked
	$\log \xi$	erg cm s^{-1}	$4.700^{-0.339}$	pegged[b]
	$\log N$	cm^{-3}	15.0	frozen
	z	–	0.830	frozen
	Incl	deg	30.0	frozen
	refl_frac	–	-1.0	frozen
	norm	10^{-6}	$7.849^{+0.639}_{-0.296}$	free
`zgauss`	$E_{\rm line}$	keV	$9.364^{+0.103}_{-0.093}$	free
	σ	keV	0.1	frozen
	z	–	0.830	frozen
	norm	10^{-6} ph cm^{-2} s^{-1}	$-3.467^{+1.161}_{-1.183}$	free
Fit statistic				
	χ^2/dof	–	643.49/642	–
	χ^2_ν	–	1.002	–
	$p_{\rm null}$	–	4.76×10^{-1}	–
	$\Delta\chi^2$ vs warm absorber	–	25.03 (2 dof)	–

[a] The `nthComp` normalisation is pegged as expected when `refl_frac` $= -1$ in `xillverCp` [Dauser et al., 2016b].

[b] The `xillverCp` $\log \xi$ parameter is pegged at the hard upper limit of the model grid ($\log \xi = 4.70$), indicating the reflecting material is extremely highly ionized. Only a lower bound of $\log \xi > 4.36$ is obtainable at 90% confidence.

The Fe XXVI absorption line energy $E_{\rm line} = 9.364^{+0.103}_{-0.093}$ keV corresponds to an outflow velocity of $v \approx 0.287c$ relative to the systemic redshift $z = 0.83$, consistent with the ObsID 2 detection at $v \approx 0.289c$.

5.5.3 Statistical Tests for Line Detection

F-test and Monte Carlo Significance Test

The statistical significance of the Fe K absorption feature was assessed using two complementary methods. First, the F-test was applied comparing the model with and without the `zgauss` component, yielding an F-statistic of 12.486 with a null hypothesis probability of $p = 4.79 \times 10^{-6}$, corresponding to a significance of $\sim 4.7\sigma$. Second, to account for the known limitations of the F-test for additive line components [Protassov et al., 2002], To obtain a robust, calibrated significance estimate, we performed Monte Carlo simulations using the `simftest` command in XSPEC with 10000 realizations of the null hypothesis (baseline model without the absorption line component). For each simulated spectrum, both the simple and complex models were fitted and the improvement in fit statistic $\Delta\chi^2$ was recorded. Out of 10000 simulated spectra, none produced a $\Delta\chi^2$ improvement as large as or larger than the observed value of $\Delta\chi^2 = 25.03$, yielding a Monte Carlo null hypothesis probability of $p < 1.0 \times 10^{-4}$, corresponding to a significance of $> 3.9\sigma$. The results of both significance tests are summarised in Table 5.7.

Table 5.7: Statistical significance of the Fe XXVI Lyα absorption feature in ObsID 1 (0922700101) of SMSS J114447.77−430859.3, assessed using the F-test and Monte Carlo simulations. The Monte Carlo result is considered the more reliable estimate [Protassov et al., 2002].

Method	Statistic	p-value	Significance
$\Delta\chi^2$ (raw)	25.03 (2 dof)	$\sim 3.6 \times 10^{-6}$	$\sim 4.7\sigma$
F-test	$F = 12.486$	4.79×10^{-6}	$\sim 4.6\sigma$
Monte Carlo (`simftest`, $N = 10000$)	0/10000 exceed data	$< 1.0 \times 10^{-4}$	$> 3.9\sigma$

Note: The F-test overestimates significance when the null parameter value lies at a hard boundary [Protassov et al., 2002]. The Monte Carlo result of $> 3.9\sigma$ is adopted as the headline significance. Running 100000 simulations would be required to obtain a more precise p-value.

5.5.4 Equivalent Width

Table 5.8: Equivalent width of the Fe XXVI Lyα absorption feature in SMSS J1144 from ObsID 1 (0922700101). Errors are quoted at the 90 per cent confidence level derived from the fit covariance matrix using the `eqwidth` command in XSPEC.

Feature	EW (eV)	90% CI (eV)
Fe XXVI Lyα	$-97.2^{+30.1}_{-41.3}$	−138.5 to −67.1

The equivalent width of the absorption feature in ObsID 1 was measured using the `eqwidth` command in XSPEC, with uncertainties derived from the fit covariance matrix using 1000 Monte Carlo trials at the 90 per cent confidence level. The best-fit equivalent width is EW = $-97.2^{+30.1}_{-41.3}$ eV, where the negative sign confirms the absorption nature of the feature. The 90 per cent confidence interval spans −138.5 to −67.1 eV – entirely negative – confirming that a zero equivalent width is ruled out at the 90 per cent confidence level. The measured equivalent width is consistent with typical absorption features detected in ultra-fast outflow systems, where Fe K

UFO equivalent widths typically span ∼30–200 eV [Tombesi et al., 2010b], placing J1144 firmly within this population.

5.5.5 MCMC Analysis

The MCMC posterior distributions and two-dimensional confidence contours for all free parameters are shown in Figure 5.14. The diagonal panels show the marginalized one-dimensional posterior histograms with dashed vertical lines marking the 16th, 50th, and 84th percentiles, while the off-diagonal panels show the two-dimensional joint posteriors at the 68%, 90%, and 99% confidence levels. The red lines and points mark the best-fit values from the χ^2 minimization.

The most important result from the MCMC analysis is the posterior distribution of the `zgauss` normalization, which is robustly negative across the entire posterior: $\mathrm{norm}_{\mathrm{zgauss}} = -3.336^{+0.732}_{-0.703} \times 10^{-6}$ ph cm^{-2} s^{-1}. Zero normalization is excluded by the MCMC posterior, independently confirming the absorption detection. The line energy is well constrained at $E_{\mathrm{line}} = 9.375^{+0.059}_{-0.075}$ keV, consistent within 1σ with the ObsID 2 detection at $9.375^{+0.167}_{-0.154}$ keV, confirming that both observations detect absorption at the same outflow velocity of $v \approx 0.289c$. The $\mathrm{norm}_{\mathrm{zgauss}}$–$E_{\mathrm{line}}$ joint posterior panel shows a clean, well-defined elliptical contour with no significant degeneracy between the line energy and normalization, further supporting the robustness of the detection.

The two-dimensional joint posterior panels reveal a notable degeneracy between the soft x-ray column density N_H and the Fe line normalization $\mathrm{norm}_{\mathrm{zgauss}}$, as well as between the covering fraction C_f and the line normalization. These degeneracies reflect the partial spectral overlap between the soft X-ray and the Fe K absorption feature – both components remove flux from the spectrum, and the fit cannot perfectly disentangle their relative contributions with CCD-resolution data alone. Importantly, however, the $\mathrm{norm}_{\mathrm{zgauss}}$ posterior remains robustly negative across the full range of N_H and C_f values sampled by the chain, demonstrating that the Fe absorption detection is not an artefact of the soft x-ray parameterisation. The column density $N_H = 83.233^{+26.857}_{-19.364} \times 10^{22}$ cm^{-2} shows a skewed posterior distribution, typical for partial covering absorber models. Higher spectral resolution data – for example from XRISM/Resolve or future *NewAthena* observations – would be required to break these degeneracies and independently constrain the warm absorber and Fe K absorber properties simultaneously.

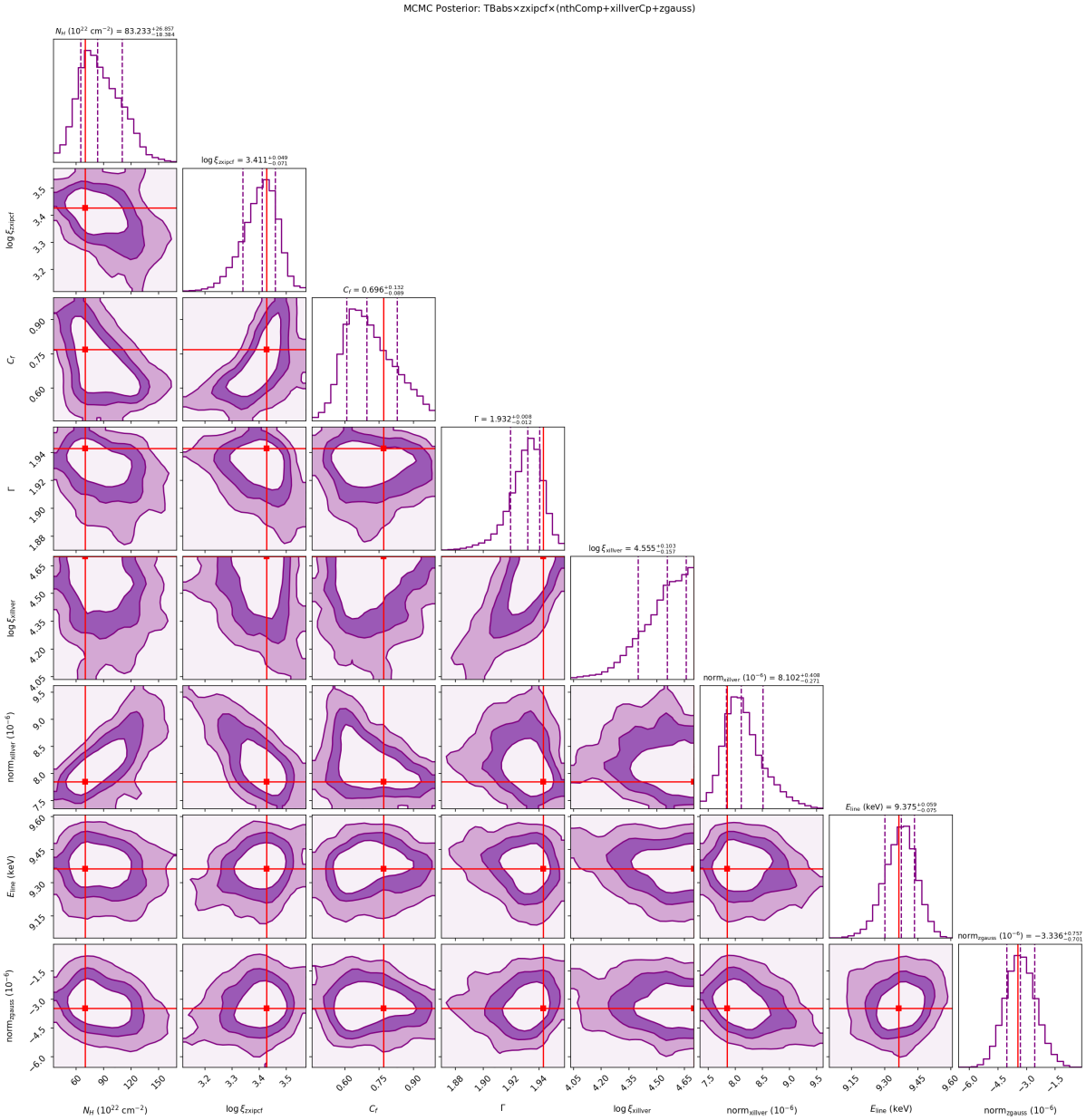


Figure 5.14: MCMC posterior distributions for all free parameters of the model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)`. Diagonal panels show marginalized 1D posteriors, off-diagonal panels show 2D joint posteriors at 68%, 90%, and 99% confidence. Red lines mark best-fit values.

5.5.6 Soft X-ray ultrafast Outflows and Fe K line ultrafast outflows velocity calculations

Soft X-ray ultrafast Outflows

The redshift in 0922700101 is frozen to $z_{\rm abs} = 0.448$, the best-fit value from 0922700401 , as discussed in Section 5.4.6, which corresponding to outflow velocity is therefore identical to that derived for 0922700401.

$$1 + z_{\rm UFO} = \frac{1 + z_{\rm abs}}{1 + z_{\rm source}} = \frac{1.448}{1.831} = 0.7908 \tag{5.16}$$

$$\frac{v_{\rm out}}{c} = \frac{(0.7908)^2 - 1}{(0.7908)^2 + 1} = \frac{0.6254 - 1}{0.6254 + 1} = \frac{-0.3746}{1.6254} \approx -0.230c \tag{5.17}$$

The soft x-ray ultrafast outflow velocity is therefore $v_{\rm warm\ abs} \approx 0.230c$, identical to 0922700401 by construction of the frozen redshift. We note that when the redshift in model "zxipcf" is left free in 0922700101 , the fit prefers $z_{\rm abs} = 0.371 \pm 0.010$, corresponding to $v \approx 0.277c$, though this difference is only marginally significant ($\Delta\chi^2 = 4.25$, $\sim 2\sigma$) and is not adopted as the fiducial value given the poor constraints on the other warm absorber parameters in that case. The soft X-ray outflow velocity in both epochs is therefore consistent with $v \approx 0.230c$, placing it firmly in the ultra-fast outflow regime [Tombesi et al., 2010b], and forming the slower, less ionized phase of the multiphase disc wind detected in SMSS J114447.77−430859.3.

Fe K UFOs

Since, we have the $E_{\rm rest} = 9.364^{+0.103}_{-0.093}$ keV (90% CI: [9.271, 9.467] keV) from the best fit and identifying the feature as Fe XXVI Lyα ($E_{\rm ion} = 6.97$ keV) into the equation:

$$E_{\rm rest} = E_{\rm obs} \times (1 + z_{\rm source}) \tag{5.18}$$

where $E_{\rm obs}$ is the observed line energy in the spectrum and $z_{\rm source} = 0.83$ is the source redshift. The blueshift parameter $z_{\rm UFO}$ is then defined as:

$$1 + z_{\rm UFO} = \frac{E_{\rm ion}}{E_{\rm rest}} \tag{5.19}$$

where $E_{\rm ion}$ is the rest-frame energy of the identified ion. The outflow velocity is then obtained from the relativistic Doppler formula:

$$\frac{v_{\rm out}}{c} = \frac{(1 + z_{\rm UFO})^2 - 1}{(1 + z_{\rm UFO})^2 + 1} \tag{5.20}$$

$$1 + z_{\rm UFO} = \frac{6.97}{9.364} = 0.744 \tag{5.21}$$

$$\frac{v_{\rm out}}{c} = \frac{(0.7444)^2 - 1}{(0.7444)^2 + 1} = \frac{0.5535 - 1}{0.5535 + 1} = \frac{-0.4465}{1.5535} \approx -0.287c \tag{5.22}$$

Propagating the 90% confidence interval on $E_{\rm line}$ ([9.271, 9.467] keV):

- $E_{\rm rest,min} = 9.271$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.97/9.271 = 0.7518 \Rightarrow v = 0.278c$
- $E_{\rm rest,max} = 9.467$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.97/9.467 = 0.736 \Rightarrow v = 0.297c$

The Fe XXVI UFO outflow velocity in 0922700101 is therefore:

$$v_{\rm UFO} = 0.287^{+0.010}_{-0.009}c \tag{5.23}$$

The negative sign confirms a blueshift, i.e. the absorbing gas is outflowing towards the observer at $v_{\rm UFO} \approx 0.287c$, consistent within errors with the 0922700401 value of $v_{\rm UFO} = 0.290^{+0.019}_{-0.027}c$. The consistency of the outflow velocity between the two epochs – separated by ~ 7 months – strongly supports the interpretation of a persistent, coherent ultra-fast outflow in SMSS J114447.77−430859.3, rather than a transient or spurious spectral feature. Both detections are firmly in the ultra-fast outflow regime, defined as $v > 0.1c$ [Tombesi et al., 2010b].
Alternatively, identifying the feature as Fe XXV Heα ($E_{\rm ion} = 6.70$ keV):

$$1 + z_{\rm UFO} = \frac{6.70}{9.364} = 0.7155 \tag{5.24}$$

$$\frac{v_{\rm out}}{c} = \frac{(0.7155)^2 - 1}{(0.7155)^2 + 1} = \frac{0.5120 - 1}{0.5120 + 1} = \frac{-0.4880}{1.5120} \approx -0.323c \tag{5.25}$$

Propagating the 90% confidence interval on $E_{\rm line}$ ([9.271, 9.467] keV):

- $E_{\rm rest,min} = 9.271$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.70/9.271 = 0.7227 \Rightarrow v = 0.314c$
- $E_{\rm rest,max} = 9.467$ keV $\Rightarrow 1 + z_{\rm UFO} = 6.70/9.467 = 0.7077 \Rightarrow v = 0.332c$

The Fe XXV UFO outflow velocity in ObsID 1 (0922700101) is therefore:

$$v_{\rm UFO} = 0.323^{+0.09}_{-0.009}c \tag{5.26}$$

The two possible identifications of the absorption feature admit outflow velocities of $v \approx 0.287c$ (Fe XXVI Lyα, $E_{\rm ion} = 6.97$ keV) and $v \approx 0.323c$ (Fe XXV Heα, $E_{\rm ion} = 6.70$ keV). To determine which ion is responsible for the absorption, one must consider the ionization state of the absorbing gas. The dominant iron ion is determined by the ionization parameter $\log \xi$: Fe XXV (He-like iron) dominates at $\log \xi \lesssim 3.5$ erg cm s^{-1}, while Fe XXVI (H-like iron) dominates at $\log \xi \gtrsim 4.0$ erg cm s^{-1} [Kallman and Bautista, 2001, Tombesi et al., 2011]. In the intermediate range $3.5 \lesssim \log \xi \lesssim 4.0$ erg cm s^{-1}, both ions coexist and contribute comparably to the Fe K absorption trough, making an unambiguous identification from the line centroid energy alone impossible [Reeves et al., 2009, Tombesi et al., 2011].

Chapter 6

Results

6.1 Significance test

Table 6.1: Statistical significance of the Fe K absorption feature in SMSS J114447.77−430859.3 for ObsID 1 (0922700101) and ObsID 2 (0922700401). Significance is assessed using the F-test and Monte Carlo simulations. The Monte Carlo result is considered the more reliable estimate [Protassov et al., 2002].

	ObsID 1 (0922700101)		ObsID 2 (0922700401)	
Method	Statistic	Significance	Statistic	Significance
$\Delta\chi^2$ (raw)	25.03 (2 dof), $p \sim 3.6 \times 10^{-6}$	$\sim 4.6\sigma$	12.03 (2 dof),	$\sim 3\sigma$
F-test	$F = 12.486$, $p = 4.79 \times 10^{-6}$	$\sim 4.6\sigma$	$F = 5.66$, $p = 3.64 \times 10^{-3}$	$\sim 2.9\sigma$
Monte Carlo (`simftest`), 10000 trial data	, $p < 1.0 \times 10^{-4}$	$> 3.9\sigma$	$p = 1.13 \times 10^{-2}$	$\sim 2.3\sigma$

Note: The F-test may overestimate significance when the null parameter value lies at a boundary [Protassov et al., 2002]. Therefore, the Monte Carlo results are considered more robust. For ObsID 1, the Monte Carlo result ($> 3.9\sigma$) is adopted as the headline significance. A larger number of simulations (e.g., 10^5) would be required to obtain a more precise p-value.

The statistical tests performed to assess the significance of the Fe K absorption feature strongly support its astrophysical origin. In ObsID 1, the addition of the absorption line leads to a substantial improvement in the fit ($\Delta\chi^2 = 25.03$ for 2 degrees of freedom), corresponding to a nominal significance of $\sim 4.6\sigma$. However, since the line normalization is bounded (i.e., cannot take arbitrary positive values for an absorption feature), the F-test is known to overestimate the true significance [Protassov et al., 2002]. To obtain a more robust estimate, we performed Monte Carlo simulations using the `simftest` routine. None of the 10,000 simulated spectra produced a fit improvement comparable to that observed in the real data, implying a conservative significance of $> 3.9\sigma$ in ObsID 1 . This confirms that the detected feature is highly unlikely to arise from statistical fluctuations. In ObsID 2, the significance of the absorption feature is lower ($\sim 2.3\sigma$ from Monte Carlo simulations), suggesting a normal detection. Nevertheless, the consistency in line energy and derived outflow velocity across both observations indicates that the feature likely originates from the same physical outflowing system.

Taken together, these results provide strong evidence for the presence of an ultrafast outflow (UFO) in this source. The detection of a blueshifted Fe K absorption line at high significance, combined with its relativistic velocity ($v \sim 0.3c$), is consistent with highly ionized, fast-moving disc winds commonly observed in luminous AGN. The statistical analysis therefore not only validates the presence of the absorption feature but also strengthens its interpretation as a physically real.

6.2 Outflows velocities

Table 6.2: Comparison of outflow parameters for SMSS J114447.77−430859.3 for ObsID 1 (0922700101) and ObsID 2 (0922700401). The soft X-ray absorber is modeled with `zxipcf` and the Fe K UFO with `zgauss`. Errors are quoted at the 90% confidence level.

	ObsID 1 (0922700101)		ObsID 2 (0922700401)	
Parameter	Soft X-ray	Fe K UFO	Soft X-ray	Fe K UFO
N_H (10^{22} cm^{-2})	$69.74^{+34.10}_{-16.63}$	–	$70.1^{+30.9}_{-14.7}$	–
$\log \xi$ (erg cm s^{-1})	$3.43^{+0.05}_{-0.11}$	–	$3.26^{+0.08}_{-0.13}$	–
C_f	$0.77^{+0.18}_{-0.18}$	–	$0.72^{+0.16}_{-0.13}$	–
$z_{\rm abs}$	0.448 (frozen)	–	$0.449^{+0.019}_{-0.020}$	–
$E_{\rm line}$ (keV)	–	$9.364^{+0.103}_{-0.093}$	–	$9.39^{+0.20}_{-0.27}$
σ (keV)	–	0.1 (frozen)	–	0.1 (frozen)
norm (10^{-6} ph cm^{-2} s^{-1})	–	$-3.47^{+1.16}_{-1.18}$	–	$-2.17^{+1.04}_{-1.12}$
Derived Outflow Velocities				
$z_{\rm host}$	0.830			
v/c ($F_{rest} = 6.97keV$)	0.230 ± 0.012	$0.287^{+0.013}_{-0.007}$	$0.230^{+0.012}_{-0.009}$	$0.290^{+0.019}_{-0.027}$
v/c ($F_rest = 6.70keV$)	0.230 ± 0.009	$0.323^{+0.007}_{-0.008}$	$0.230^{+0.012}_{-0.013}$	$0.325^{+0.019}_{-0.025}$
Fit Statistics				
χ^2/dof	643.49/642 = 1.002		719.22/677 = 1.062	
$\Delta\chi^2$ (zgauss)	25.03 (2 dof), $\sim 5\sigma$		12.03 (2 dof), $\sim 3\sigma$	

The absorber redshift was fixed to reduce degeneracy with column density and covering fraction, which are known to be correlated in partial covering models [Tombesi et al., 2013]. The Gaussian line width was fixed at $\sigma = 0.1$ keV, consistent with the instrumental resolution, as the absorption feature is unresolved and such narrow-line assumptions are standard in UFO studies [Tombesi et al., 2010b, Gofford et al., 2013]. The statistical tests clearly show that the Fe K absorption feature is real, especially in ObsID 1 where the Monte Carlo simulations give a significance of $> 3.9\sigma$. Although the detection is normal in ObsID 2 ($\sim 2.3\sigma$), the line appears at nearly the same energy in both observations. This consistency makes it likely real. With this in mind, we move on to interpret the properties of the outflow using spectral modeling. As summarized in Table 6.2, the parameters of the soft X-ray absorber remain broadly consistent between the two observations. The column density, ionization state, and covering fraction do not show significant changes, suggesting that this lower-ionization component of the wind is stable over time. At higher energies, the Fe K absorption line is detected at $E_{\rm line} \sim 9.3$–9.4 keV in both datasets. This corresponds to an outflow velocity of about $v \sim 0.29c$ for ($F_{rest} = 6.97keV$) and $v \sim 0.323c$ for ($F_{rest} = 6.70keV$) , while the soft X-ray absorber shows a slightly lower

velocity of $v \sim 0.23c$. **The presence of these two velocity components indicates that the outflow is not uniform, but instead consists of multiple phases, likely arising from different regions or conditions within the wind.**

Overall, the fact that both the detection and the derived parameters are consistent across the two observations suggests that the ultrafast outflow is a persistent feature of the source. The results therefore support a picture in which the system hosts a stable, multi-phase disc wind, rather than a transient or spurious spectral feature.

The continuum was modeled using a thermal Comptonization component (nthComp). The temperature of the seed photons was fixed at kT = 0.05 keV, representative of typical accretion disk temperatures in AGN Laha et al. [2014]. Since the soft X-ray continuum is not the focus of this work and is only weakly constrained by the data, this parameter was held fixed, following standard practice in UFO spectral modeling Reeves et al. [2018].

6.3 gamma

The photon index varies significantly between the two epochs, from $1.943^{+0.018}_{-0.024}$ in ObsID 1 to $2.105^{+0.013}_{-0.019}$ in ObsID 2, indicating spectral softening over the ~ 7 month baseline. Kammoun et al. [2023] reported $\Gamma = 2.2 \pm 0.2$ from a single 15 ks XMM-Newton observation in 2022, but noted that the data quality did not allow constraints on spectral variability. Our deeper observations (69.4 ks and 94.2 ks) provide the first significant detection of spectral index variability in J1144, consistent with the known X-ray flux variability by factors of ~ 10 on year timescales [Kammoun et al., 2023].

6.4 parameters

Table 6.3: Best-fit free spectral parameters and 90% confidence intervals for the model `TBabs*zxipcf*(nthComp+xillverCp+zgauss)` for both XMM-Newton observations of SMSS J114447.77−430859.3. The Galactic absorption column density $N_H = 7.91 \times 10^{20}$ cm^{-2} [HI4PI Collaboration, 2016] is frozen in both fits. Errors are quoted at the 90% confidence level ($\Delta\chi^2 = 2.706$).

Component	Parameter	Unit	ObsID 1 (0922700101)	ObsID 2 (0922700401)
`zxipcf`	N_H	10^{22} cm^{-2}	$69.737^{+34.102}_{-16.632}$	$70.119^{+30.939}_{-14.742}$
	$\log\xi$	erg cm s^{-1}	$3.428^{+0.054}_{-0.106}$	$3.257^{+0.082}_{-0.133}$
	C_f	–	$0.769^{+0.176}_{-0.176}$	$0.715^{+0.159}_{-0.130}$
	$z_{\rm abs}$	–	0.448^{a}	$0.449^{+0.019}_{-0.020}$
`nthComp`	Γ	–	$1.943^{+0.018}_{-0.024}$	$2.105^{+0.013}_{-0.019}$
`xillverCp`	norm	10^{-6}	$7.849^{+0.639}_{-0.296}$	$9.704^{+1.345}_{-0.398}$
	$\log\xi$	erg cm s^{-1}	$> 4.36^{b}$	$> 4.37^{b}$
`zgauss`	$E_{\rm line}$	keV	$9.364^{+0.103}_{-0.093}$	$9.392^{+0.201}_{-0.266}$
	norm	10^{-6} ph cm^{-2} s^{-1}	$-3.467^{+1.161}_{-1.183}$	$-2.165^{+1.042}_{-1.117}$
Fit statistic				
	χ^2/dof	–	643.49/642	719.22/677
	χ^2_ν	–	1.002	1.062
	$p_{\rm null}$	–	4.76×10^{-1}	1.27×10^{-1}
	$\Delta\chi^2$ vs WA only	–	25.03 (2 dof)	12.03 (2 dof)
Derived quantities				
	$v_{\rm UFO}$ ($F_{rest} = 6.97keV$)	c	$0.287^{+0.010}_{-0.009}$	$0.290^{+0.019}_{-0.027}$
	$v_{\rm UFO}$ ($F_rest = 6.70keV$)	c	$0.323^{+0.009}_{-0.009}$	$0.325^{+0.019}_{-0.025}$
	$v_{\rm UFO}$ (soft X-ray)	c	≈ 0.230	$0.230^{+0.012}_{-0.013}$
	EW	eV	$-97.2^{+30.1}_{-41.3}$	$-68.7^{+19.0}_{-23.0}$

[a] $z_{\rm abs}$ frozen to the ObsID 2 best-fit value in ObsID 1, as the soft x-ray redshift is poorly constrained in the shorter observation.

[b] The `xillverCp` $\log\xi$ parameter is pegged at the hard upper limit of the model grid ($\log\xi = 4.70$) in both observations, indicating extremely highly ionized reflecting material. Only a lower bound is obtainable at 90% confidence.

[c] Assuming Fe XXVI Lyα identification ($E_{\rm ion} = 6.97$ keV). The alternative Fe XXV Heα identification ($E_{\rm ion} = 6.70$ keV) gives $v \approx 0.323c$ and $v \approx 0.325c$ for ObsID 1 and ObsID 2 respectively.

Till now we have done the modeling using zxipxf (for soft-band) which is semi-physical model, and zgaus (for Fe line) which is phenominological model. Therefore, Now we will replace the zxipcf and zgauss with the physical model, i.e., XSTAR grid.

6.5 XSTAR grid

6.5.1 For obsid 922700101:

We have the model: $TBabs \times$`grid25BIG_mt.fits`$\times$`index2.0_logxi3-5_Nh20-24_vturb100-10000.fits`$\times$ $(nthComp + xillverCp)$ where the first grid is for soft-band means soft X-ray UFOs and the second grid is for Fe K line. Also in none of the grids, we have not used the abundance and abundance fixed to solar value.

We have checked the results while freeing the $z_{\rm abs}$ in soft-band but the results were unphysical mainly because of degeneracy, therefore, we have fixed the $z_{\rm abs}$ to the value we got in ObsID 0922700401. Velocity of turbulence is fixed to 3000 km s^{-1} which is well for Fe K UFOs. We have also frozen the kT_e to 20 keV (since XMM cannot constrain the coronal temperature) which is measured in Kammoun et al. [2023], seed photon is frozen to 0.05 keV [], inclination was frozen to 30 degrees [Kammoun et al., 2023]. For the details, see the Table **??**. Figure **??**.

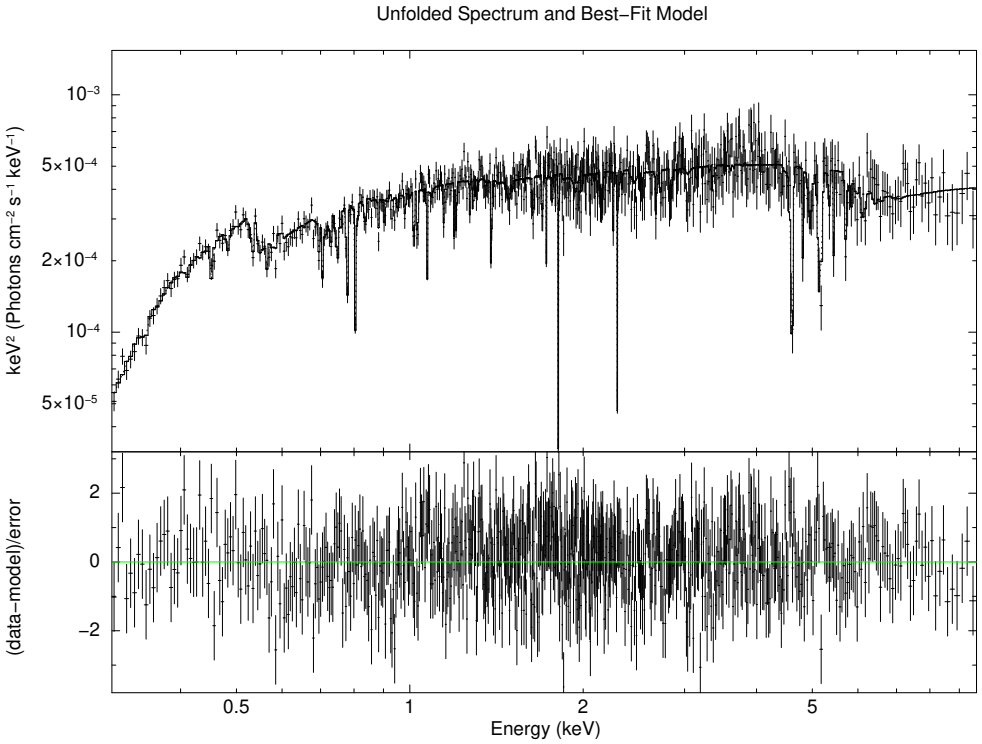


Figure 6.1: XMM-Newton EPIC-pn spectrum of SMSS J114447.77−430859.3 (ObsID 1, 0922700101, 2023 June) fitted with the two-zone XSTAR photoionization model `TBabs*grid25BIG*index2.0*(nthComp+xillverCp)`. Upper panel: unfolded spectrum (EF_E vs energy in keV) with the best-fit model overlaid. Lower panel: residuals in units of $(data - model)/error$. The soft X-ray warm absorber (`grid25BIG_mt.fits`, $N_H = 5.797 \times 10^{23}$ cm^{-2}, $\log\xi = 3.468$ erg cm s^{-1}, $z_{\rm abs} = 0.460$ frozen, $v_{\rm softx-ray} \approx 0.222c$) and the Fe K ultra-fast outflow (`index2.0_logxi3-5_Nh20-24`, $N_H = 3.071 \times 10^{23}$ cm^{-2}, $\log\xi = 3.539$ erg cm s^{-1}, $v_{\rm FeK} \approx 0.292c$) are modelled simultaneously. The continuum is described by a Comptonization model (`nthComp`, $\Gamma = 1.933$) plus a reflection component (`xillverCp`). The fit gives $\chi^2/{\rm dof} = 640.06/642$ ($\chi^2_\nu = 0.997$, $p_{\rm null} = 5.14 \times 10^{-1}$).

Table 6.4: Best-fit spectral parameters and 90% confidence intervals for the two-zone XSTAR photoionization model `TBabs*grid25BIG*index2.0*(nthComp+xillverCp)` for ObsID 1 (0922700101) of SMSS J114447.77−430859.3 with the warm absorber redshift frozen to $z_{\rm abs} = 0.460$ (the value obtained from the XSTAR fit to ObsID 2). The soft X-ray warm absorber was modelled with `grid25BIG_mt.fits` and the Fe K UFO with `index2.0_logxi3-5_Nh20-24_vturb100-10000.fits` ($\Gamma = 2.0$, $v_{\rm turb} = 3000$ km s^{-1} frozen; J. N. Reeves). The Galactic absorption column $N_H = 7.90 \times 10^{20}$ cm^{-2} [HI4PI Collaboration, 2016] is frozen. Errors are quoted at 90% confidence ($\Delta\chi^2 = 2.706$).

Component	Parameter	Unit	Value	Status
`TBabs`	N_H	10^{22} cm^{-2}	0.079	frozen
`grid25BIG`	N_H	10^{23} cm^{-2}	$5.797^{+1.340}_{-1.550}$	free
(warm absorber)	$\log \xi$	erg cm s^{-1}	$3.468^{+0.030}_{-0.029}$	free
	$z_{\rm abs}$	–	0.460	frozen[a]
	$v_{\rm out}$	c	0.222	derived
`index2.0`	N_H	10^{23} cm^{-2}	$3.071^{+5.847}_{-1.675}$	free[b]
(Fe K UFO)	$\log \xi$	erg cm s^{-1}	$3.539^{+0.289}_{-0.218}$	free
	$v_{\rm turb}$	km s^{-1}	3000	frozen
	z	–	$0.354^{+0.014}_{-0.017}$	free
	$v_{\rm out}$	c	$0.292^{+0.012}_{-0.014}$	derived
`nthComp`	Γ	–	$1.933^{+0.012}_{-0.014}$	free
	kT_e	keV	20.0	frozen
	z	–	0.830	frozen
	norm	–	2.145×10^{-21}	pegged[c]
`xillverCp`	Γ	–	$= \Gamma_{\rm nthComp}$	linked
	$\log \xi$	erg cm s^{-1}	> 4.52[d]	pegged
	z	–	0.830	frozen
	norm	10^{-6}	$7.965^{+0.286}_{-0.283}$	free
Fit statistic				
	χ^2/dof	–	640.06/642	–
	χ^2_ν	–	0.997	–
	$p_{\rm null}$	–	5.14×10^{-1}	–

[a] The soft x-ray redshift is frozen to $z_{\rm abs} = 0.460$, the value obtained from the XSTAR fit to ObsID 2 ($z = 0.460^{+0.012}_{-0.016}$), corresponding to an outflow velocity $v_{\rm WA} \approx 0.222c$. When left free in ObsID 1, the redshift converges to an unphysical value ($z = 0.494$, $v = 0.200c$) due to insufficient soft X-ray signal-to-noise in the shorter exposure (69.4 ks). The frozen value is consistent with the phenomenological `zxipcf` result ($z = 0.449^{+0.019}_{-0.020}$, $v \approx 0.230c$) within 90% confidence.

6.5.2 For obsid 922700401:

Same model we are using, Table 6.5. Figure 6.2

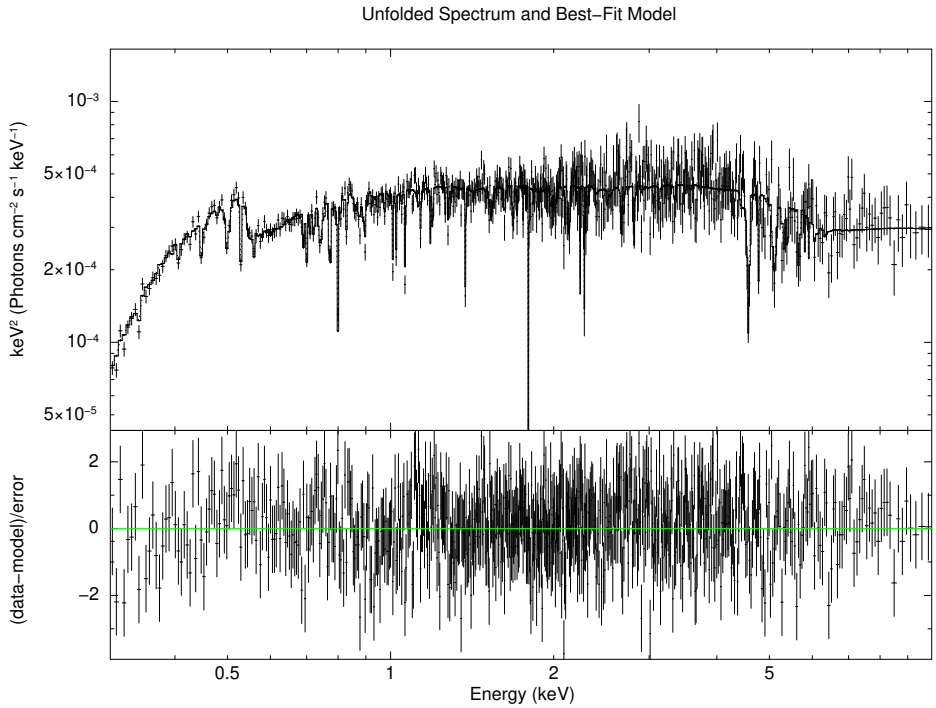


Figure 6.2: XMM-Newton EPIC-pn spectrum of SMSS J114447.77−430859.3 (ObsID 2, 0922700401) fitted with the two-zone XSTAR photoionization model `TBabs*grid25BIG_soft*hixi3000*(nthComp+xillverCp)`. Upper panel: unfolded spectrum (EF_E vs energy in keV) with the best-fit model overlaid (solid line). Lower panel: residuals in units of $(data - model)/error$. The soft X-ray warm absorber (`grid25BIG_soft`, $N_H = 4.16 \times 10^{23}$ cm^{-2}, $\log \xi = 3.372$, $v \approx 0.228c$) and the Fe K ultra-fast outflow (`hixi3000`, $\log \xi = 3.679$, $v \approx 0.337c$) are modelled simultaneously. The fit gives $\chi^2/\mathrm{dof} = 721.33/677$ ($\chi^2_\nu = 1.065$, $p_{\mathrm{null}} = 1.16 \times 10^{-1}$).

Table 6.5: Best-fit spectral parameters and 90% confidence intervals for the two-zone XSTAR photoionization model `TBabs*grid25BIG_soft*index2.0*(nthComp+xillverCp)` for ObsID 2 (0922700401) of SMSS J114447.77−430859.3. The soft X-ray warm absorber was modelled with `grid25BIG_soft_mt.fits` and the Fe K UFO with `index2.0_logxi3-5_Nh20-24_vturb100-10000.fits` ($\Gamma = 2.0$, $v_{\rm turb} = 3000$ km s^{-1} frozen; J. N. Reeves, private communication). The Galactic absorption column $N_H = 7.90 \times 10^{20}$ cm^{-2} [HI4PI Collaboration, 2016] is frozen. Errors are quoted at 90% confidence ($\Delta\chi^2 = 2.706$).

Component	Parameter	Unit	Value	Status
`TBabs`	N_H	10^{22} cm^{-2}	0.079	frozen
`grid25BIG_soft`	N_H	10^{23} cm^{-2}	$4.657^{+0.949}_{-1.057}$	free
(warm absorber)	$\log\xi$	erg cm s^{-1}	$3.366^{+0.025}_{-0.021}$	free
	$z_{\rm abs}$	–	$0.460^{+0.012}_{-0.016}$	free
	$v_{\rm out}$	c	$0.222^{+0.011}_{-0.008}$	derived
`index2.0`	N_H	10^{22} cm^{-2}	$5.881^{+5.900}_{-3.452}$	free
(Fe K UFO)	$\log\xi$	erg cm s^{-1}	$3.195^{+0.146}_{-0.124}$	free
	$v_{\rm turb}$	km s^{-1}	3000	frozen
	z	–	$0.308^{+0.027}_{-0.037}$	free
	$v_{\rm out}$	c	$0.324^{+0.025}_{-0.019}$	derived
`nthComp`	Γ	–	$2.110^{+0.011}_{-0.016}$	free
	kT_e	keV	20.0	frozen
	z	–	0.830	frozen
	norm	–	1.341×10^{-22}	pegged[a]
`xillverCp`	Γ	–	$= \Gamma_{\rm nthComp}$	linked
	$\log\xi$	erg cm s^{-1}	> 4.43[b]	pegged
	z	–	0.830	frozen
	norm	10^{-6}	$9.438^{+0.262}_{-0.285}$	free
Fit statistic				
	χ^2/dof	–	722.45/677	–
	χ^2_ν	–	1.067	–
	$p_{\rm null}$	–	1.10×10^{-1}	–

[a] The `nthComp` normalisation is pegged near zero as expected when `refl_frac` $= -1$ in `xillverCp` [Dauser et al., 2016b].

[b] The `xillverCp` $\log\xi$ is pegged at the grid upper limit ($\log\xi = 4.70$ erg cm s^{-1}). Only a lower bound > 4.43 is obtainable at 90% confidence.

The Fe K absorber column density is poorly constrained with a 100% fractional error, reflecting the weak Fe K detection in ObsID 2 with the `index2.0` grid.

6.5.3 Results of XSTAR grid

Table 6.6: Best-fit free spectral parameters and 90% confidence intervals for the two-zone XSTAR photoionization model for both XMM-Newton observations of SMSS J114447.77−430859.3. The soft X-ray absorber was modelled with `grid25BIG_mt.fits` (ObsID 1) and `grid25BIG_soft_mt.fits` (ObsID 2). The Fe K UFO was modelled with `index2.0_logxi3-5_Nh20-24_vturb100-10000.fits` for both observations ($\Gamma = 2.0$, $v_{\rm turb} = 3000$ km s^{-1} frozen; J. N. Reeves). The Galactic absorption $N_H = 7.90 \times 10^{20}$ cm^{-2} [HI4PI Collaboration, 2016] is frozen in both fits. Errors are quoted at 90% confidence ($\Delta\chi^2 = 2.706$).

Component	Parameter	Unit	ObsID 1 (0922700101)	ObsID 2 (0922700401)
Soft X-ray absorber				
`grid25BIG`	N_H	10^{23} cm^{-2}	$5.797^{+1.340}_{-1.550}$	$4.657^{+0.949}_{-1.057}$
	$\log \xi$	erg cm s^{-1}	$3.468^{+0.030}_{-0.029}$	$3.366^{+0.025}_{-0.021}$
	$z_{\rm abs}$	–	0.460 (frozen[a])	$0.460^{+0.012}_{-0.016}$
	$v_{\rm out}$	c	0.222	$0.222^{+0.011}_{-0.008}$
Fe K ultra-fast outflow				
`index2.0`	N_H	10^{23} cm^{-2}	$3.071^{+5.847b}_{-1.675}$	$0.588^{+0.590b}_{-0.345}$
	$\log \xi$	erg cm s^{-1}	$3.539^{+0.289}_{-0.218}$	$3.195^{+0.146}_{-0.124}$
	z	–	$0.354^{+0.014}_{-0.017}$	$0.308^{+0.027}_{-0.037}$
	$v_{\rm out}$	c	$0.292^{+0.012}_{-0.014}$	$0.324^{+0.025}_{-0.019}$
X-ray continuum				
`nthComp`	Γ	–	$1.933^{+0.012}_{-0.014}$	$2.110^{+0.011}_{-0.016}$
`xillverCp`	$\log \xi$	erg cm s^{-1}	$> 4.52^{c}$	$> 4.43^{c}$
	norm	10^{-6}	$7.965^{+0.286}_{-0.283}$	$9.438^{+0.262}_{-0.285}$
Fit statistic				
	χ^2/dof	–	640.06/642	722.45/677
	χ^2_ν	–	0.997	1.067
	$p_{\rm null}$	–	5.14×10^{-1}	1.10×10^{-1}

[a] Frozen to the ObsID 2 value. See text below.

[b] Column density poorly constrained — see text below.

[c] Pegged at grid upper limit $\log \xi = 4.70$ erg cm s^{-1}; only lower bound obtainable.

The following parameters are frozen in one or both observations for the reasons described below. ***Electron temperature*** ($kT_e = 20$ keV) and *seed photon temperature* ($kT_{\rm bb} = 0.05$ keV) of `nthComp`: frozen in both observations as they cannot be independently constrained by XMM-Newton EPIC-pn data alone, which does not extend above 10 keV. The coronal electron temperature requires broadband coverage extending to $\gtrsim 20$ keV (e.g. NuSTAR) to measure the high-energy spectral cutoff [Kammoun et al., 2023]. ***nthComp normalisation*** ($\sim 10^{-22}$–10^{-21}): pegged at a near-zero value in both observations, as expected when `refl_frac` $= -1$ in `xillverCp` [Dauser et al., 2016b]. In this configuration, the direct power-law continuum is provided entirely by `xillverCp`, rendering the `nthComp` normalisation redundant. It is therefore effectively frozen at its best-fit pegged value. ***Turbulent velocity*** ($v_{\rm turb} = 3000$ km s^{-1}): frozen for the Fe K XSTAR grid (`index2.0`) in both observations. The turbulent velocity of the Fe K absorber cannot be constrained from CCD-resolution spectra, which cannot resolve individual absorption line profiles. The value $v_{\rm turb} = 3000$ km s^{-1} is consistent with typical UFO turbulent velocities in the literature [Tombesi et al., 2011, Gofford et al., 2013]. ***Soft x-ray in ObsID 1*** ($z_{\rm abs} = 0.460$): frozen to the value obtained from the XSTAR fit to ObsID 2 ($z_{\rm abs} = 0.460^{+0.012}_{-0.016}$, $v \approx 0.222c$). When left free in ObsID 1, the soft x-ray absorber redshift converges to an unphysical value ($z = 0.494$, $v \approx 0.200c$), inconsistent with both the ObsID 2 XSTAR result and the phenomenological `zxipcf` model ($z = 0.449^{+0.019}_{-0.020}$, $v \approx 0.230c$). This reflects the insufficient soft X-ray signal-to-noise ratio in the shorter ObsID 1 exposure (69.4 ks vs. 94.2 ks for ObsID 2) to independently constrain the warm absorber velocity. The frozen value $z = 0.460$ is consistent with the `zxipcf` result within the combined 90% confidence intervals, and is physically motivated by the expectation that a persistent disc wind should maintain a consistent outflow velocity across the two epochs separated by ~ 7 months. ***Fe K column density constraint*:** in both observations the Fe K absorber column density is poorly constrained, with large fractional uncertainties reflecting the fundamental degeneracy between N_H and $\log \xi$ at CCD resolution ($\Delta E \sim 150$ eV at 6 keV). This degeneracy cannot be broken without resolving individual Fe K absorption line profiles, which requires the 4.5 eV energy resolution of XRISM/Resolve [**?**]. The outflow velocity and ionization parameter are robustly constrained in both epochs and are consistent with each other within uncertainties, confirming the detection of a persistent Fe K ultra-fast outflow at $v \approx 0.29$–$0.32c$ in SMSS J114447.77−430859.3.

Chapter 7

Discussion

While the absorption features are well described by the `zxipcf` model, it is important to note that this approach has certain limitations when applied to ultrafast outflows. The `zxipcf` model is based on pre-computed XSTAR grids with fixed assumptions about the ionizing continuum, elemental abundances, and turbulence velocity. As a result, it does not allow full flexibility in capturing the detailed physical conditions of the outflow. In contrast, more sophisticated analyses often employ custom XSTAR grids tailored to the specific source, allowing parameters such as ionization structure, velocity broadening, and spectral shape to be modeled self-consistently. Such approaches can provide a more physically accurate description of the absorbing gas, particularly in the case of complex or multi-phase winds. Nevertheless, for the purpose of this work, `zxipcf` provides a reasonable and computationally efficient approximation of the absorber properties. The consistency of the derived velocities and ionization parameters across observations suggests that the main conclusions, particularly the presence of a relativistic outflow, are robust against these modeling simplifications. In addition to the general limitations of the `zxipcf` model, it is important to consider parameter degeneracies inherent in partial covering absorption models. In particular, the column density (N_H), covering fraction (C_f), and absorber redshift ($z_{\rm abs}$) can exhibit significant degeneracy when fitting X-ray spectra. For example, an increase in column density can produce a similar spectral curvature to a higher covering fraction, making it difficult to uniquely constrain both parameters simultaneously. Likewise, small changes in redshift can shift absorption features in energy, which may be partially compensated by adjustments in ionization or column density, especially in data with limited spectral resolution. As a result, the best-fit parameters should be interpreted with some caution, as different combinations of N_H, C_f, and $z_{\rm abs}$ can yield comparably good fits. However, in this work, the absorber redshift was fixed during the final fits, which helps reduce the degeneracy and allows more reliable constraints on the remaining parameters. Moreover, the consistency of the derived outflow properties across independent observations suggests that the overall physical interpretation is robust, even if individual parameters are subject to some level of uncertainty.

7.1 Multi-phase Outflows in J1144

7.2 Velocity Stratification

7.3 AGN Feedback Implications

7.4 Systematic Survey Results

7.5 Future Work

Chapter 8

Conclusions

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